ECON6101 Microeconomics for Honours Course Notes

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I use course materials from semester 1, 2025 to make most of the notes and then use 2026's materials to refine the notes.

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Part I
Economic Agents

1 Introduction to Consumer Theory

1.1 Logic

"iff" is used for definitions.

Example 1.1.1. A person x is called rational iff x reasons about their preferences and x chooses the best available alternative. This means x must satisfy both properties listed. The property "being rational" is defined by the right hand side of the iff.

"iff" also used in the logical statement "A iff B" i.e. "A if and only if B" i.e. $A \Leftrightarrow B$.

Definition 1.1.2.
- $A \Leftrightarrow B$ defined logically to hold if both $A \Rightarrow B$ and $B \Rightarrow A$ hold.
- $A \Rightarrow B$ is defined as $B \vee \neg A$.

There are two logical equivalence table.

A	B	$\neg A$	$\neg B$	$A \Rightarrow B$ $(B \vee \neg A)$	$\neg B \Rightarrow \neg A$ $(\neg A \vee \neg\neg B)$	$\neg A$	$\neg\neg B$ B
T	T	F	F	T	T	F	T
T	F	F	T	F	F	F	F
F	T	T	F	T	T	T	T
F	F	T	T	T	T	T	F

Note that the contrapositive here: $A \Rightarrow B$ is equivalent to $\neg B \Rightarrow \neg A$.

A	B	C	$A \Rightarrow B \vee C$	$B \vee C \vee \neg A$	$B \vee C$	$\neg A$	$A \wedge \neg B \Rightarrow C$	$C \vee \neg(A \wedge \neg B)$	$\neg(A \wedge \neg B)$	$(A \wedge \neg B)$	$\neg B$
T	T	T	T	T	T	F	T	T	T	F	F
T	T	F	T	T	T	F	T	T	T	F	F
T	F	T	T	T	T	F	T	T	F	T	T
T	F	F	F	F	F	F	F	F	F	T	T
F	T	T	T	T	T	T	T	T	T	F	F
F	T	F	T	T	T	T	T	T	T	F	F
F	F	T	T	T	T	T	T	T	T	F	T
F	F	F	T	T	F	T	T	T	T	F	T

Note that $A \Rightarrow (B \vee C)$ is equivalent to $(A \wedge \neg B) \Rightarrow C$.

1.2 Decision Problems and Binary Relations

Decision maker chooses from a set A of alternatives. A set mathematically is just a list of things.

Example 1.2.1. $A = \{R\&J, O, U, M\}$ (Romeo and Juliet, Origin of Species, Unbearable Lightness of Being, or Advanced Microeconomic Theory). Pick one and only one from this set.

Economics presumes you can rank alternatives and will pick one that is a best option. Maybe there is more than one best option!

Alternative x is best in A iff $x \in A$ and x is "at least as good" as y for every other $y \in A$. We can use $\succeq$ to mean "is at least as good as". Then we can write:

Definition 1.2.2. x is best in A iff $x \in A$ and $x \succeq y$ for all $y \in A$.

Note the similarity and difference between $\geq$ and $\succeq$. What do we require of $\geq$ when comparing numbers?

1. $a \geq a$ (reflexivity)
2. Either $a \geq b$ or $b \geq a$ (completeness)
3. If $a \geq b$ and $b \geq c$ then $a \geq c$ (transitivity)
4. If $a \geq b$ and $b \geq a$ then $a = b$ (anti-symmetry)

Because all four properties are satisfied for $\geq$ on $\mathbb{R}$, we say $\geq$ is a "total order". But more generally, it is an example of a binary relation (no axiom is needed!).

We want to say "bundle a is at least as good as bundle b". This is a binary relation on a set of alternatives. It is a set of ordered pairs.

Definition 1.2.3. Let X be a set. A binary relation is a subset of $X \times X$.

Re-writing "$a \succeq b$" as "$(a, b) \in \succeq$" reveals that $\succeq$ is a binary relation on the set of alternatives. This one has an interpretation: $(a, b) \in \succeq$ means "a is at least as good as b". Others may be interpretable as "is the mother of", "loves", etc. (Or nothing intuitive.)

Example 1.2.4. Suppose ranking is:

$$\{U, O\} \text{ on top, then } R\&J, \text{ then } M.$$

What is the binary relation here?

Here $A = \{U, O, R\&J, M\}$ and $\succeq \subseteq A \times A$. We have:

- $(U, O) \in \succeq$ and $(O, U) \in \succeq$ (tie at top).
- $(U, R\&J), (O, R\&J) \in \succeq$, and $(R\&J, M) \in \succeq$
- $(U, M), (O, M) \in \succeq$
- $(x, x) \in \succeq \quad \forall x \in A$ (reflexivity)

Note $a \succeq b$ is shorthand for $(a, b) \in \succeq$.

In this example, $\succeq \subset A \times A$ (a strict subset). The missing comparisons comprise the complement, $\succeq^c$. Here, $\succeq^c = \{(M, U), (M, O), (R\&J, U), (R\&J, O), (M, R\&J)\}$.

Remark. Sets use $\{\cdot\}$: order does not matter.

$$\{1, 2, 3\} = \{2, 1, 3\}$$

Relations use ordered pairs (a, b): order does matter.

$$(1, 2) \neq (2, 1).$$

Example 1.2.5. How would you represent $\geq$ as a binary relation on $\mathbb{R} = (-\infty, \infty)$?

$$\geq = \{(a, b) \in \mathbb{R} \times \mathbb{R} : a \geq b\}.$$

Another possible definition:

Definition 1.2.6. x is best in A iff $x \in A$ and no $y \in A$ is strictly better than x. Here we define strictly better than as:

$$y \succ x \iff (y \succeq x) \text{ and } \neg(x \succeq y).$$

Example 1.2.7. Suppose that the binary relation $\succeq$ is empty but A is not empty.

- What is best under Definition 1.2.2? There is no best!
- Definition 1.2.6? Every element is the best!
- Is either definition stronger? Definition 1.2.2 in this case.

Example 1.2.8. Suppose $A = \{a, b, c\}$ is the set of alternatives. Can you find a binary relation on A that does not have a best element according to Definition 1.2.2?

$$\succeq = \{(a, b)\}.$$

According to Definition 1.2.6?

$$\succeq = \{(a, b), (b, c), (c, d)\}.$$

We will use Definition 1.2.2 within this course.

Definition 1.2.9. We often are interested in whether a binary relation satisfies:

- Reflexive: $a \succeq a$ for all a.
- Transitive: $a \succeq b$ and $b \succeq c$ imply $a \succeq c$.
- Complete: for all a, b, either $a \succeq b$ or $b \succeq a$ (or both).
- Antisymmetric (for orders): if $a \succeq b$ and $b \succeq a$, then $a = b$.

Example 1.2.10 (Antisymmetry Example (Numbers))**.** For $\geq$ on numbers: If $a \geq b$ and $b \geq a$, then $a = b$.

Did previous examples (for the book example) have antisymmettric? No becasuse the books are not identical.

Did previous examples have this transitivity? Yes!

Remark. Completeness: For any $a, b \in X$, we can compare them

- $a \succeq b$ or $b \succeq a$ (possibly both).
- Allows ties (indifference).

1.3 Utility

Completeness and transitivity are enough to create a ranking (formally, a total preorder). Antisymmetry gives a strict order (no ties).

Recall from ECON3010. Let $\succeq$ be a binary relation on a finite set X. Then, $\succeq$ is complete and transitive if and only if it has a utility representation.

Thus, for finite or countable sets, a ranking is enough to be able to represent preferences with a utility function. For uncountable sets, we will need a little more - see next week's lecture.

Definition 1.3.1. A utility representation is a real-valued function $u : X \to \mathbb{R}$ such that:

$$x \succeq y \iff u(x) \geq u(y).$$

Example 1.3.2.

$$u(U) = u(O) = 2, \quad u(R\&J) = 1, \quad u(M) = 0.$$

Also Recall from ECON3010. Let X be a consumption set and $\succeq$ a complete and transitive preference relation on it. Then every non-empty and finite subset A of X has a greatest element.

Thus, with Definition 1.2.2, as long as $\succeq$ is complete and transitive on a finite set of alternatives, there will be at least one "best" alternative. Otherwise there MAY not be a best alternative.

- What is an example without completeness or transitivity where there is still a best element?

$$\text{Let } A = \{a, b, c\}. \text{ Define } \succeq = \{(a, a), (b, b), (c, c), (a, b), (a, c)\}.$$

- What is an example of a complete and transitive relation on a non-finite set without a best element?

$$\text{Take } X = \mathbb{R}. \text{ Let the binary relation be the usual "}\geq\text{": } \succeq = \{(x, y) \in \mathbb{R}^2 : x \geq y\}.$$

2 Preferences and Utility

2.1 Math Review

Definition 2.1.1. A set is a collection of objects. Subset: $S \subseteq T$ iff every element of S is in T. Empty set: $\varnothing$.

We need to distinguish three concepts:

- Set: $\{1, 2, 3\} = \{3, 2, 1\}$.
- Ordered pair: $(1, 2) \neq (2, 1)$.
- n-tuple: $(1, 2, 3) \neq (3, 2, 1)$.

The number of the subsets of a finite set with elements n is 2^n.

Example 2.1.2. $\{a, b, c\}$. There are $2^3 = 8$ subsets.

$\mathbb{R} = (-\infty, \infty)$ also has subsets: $\mathbb{R}$, $\varnothing$, $\mathbb{R}_+ = \{x \in \mathbb{R} : x \geq 0\}$, $\mathbb{R}_{++} = \{x \in \mathbb{R} : x > 0\}$, $A = \{x \in \mathbb{R} : 4 \leq x < 5\} = [4, 5)$, and $\mathbb{N} = \{1, 2, 3, \ldots\}$...

$\mathbb{R}$ is our starting point for consumer theory, with its standard properties:

- Field axioms: associative and commutative properties of $+$ and $\times$
- Order axioms: e.g. there is a binary relation $\leq$ that you are familiar with
- Least upper bound property: Each non-empty subset of $\mathbb{R}$ that has an upper bound has a least upper bound

Definition 2.1.3. x is an upper bound of $A \subset \mathbb{R}$ iff $x \geq y$ for all $y \in A$. Least upper bound: the smallest such x.

Example 2.1.4. Is 2 an upper bound for $[\sqrt{2}, 5)$? No. 20? Yes. What is the least upper bound? 5. What about $\mathbb{Q} \cap [-\sqrt{2}, \sqrt{2}]$? $\sqrt{2}$.

Suppose we have many goods. First label the goods by $1, 2, \ldots, n$ (for example, 1 is bread, 2 is milk, ...). Let the quantity of each good be x_i, and usually assume quantities cannot be negative: $x_i \geq 0$. A consumption bundle (bundle) is the n-tuple that collects all quantities:

$$\boldsymbol{x} = (x_1, \ldots, x_n).$$

Definition 2.1.5. The consumption set is the set of all feasible consumption bundles.

In this course we usually take

$$\mathbb{R}^n_+ = \underbrace{\mathbb{R}_+ \times \cdots \times \mathbb{R}_+}_{n \text{ times}}, \qquad \text{where } \mathbb{R}_+ = \{x \in \mathbb{R} : x \geq 0\}.$$

Example 2.1.6 ($n = 3$)**.** • $(2, 1, 0) \in \mathbb{R}^3_+$ (zero is allowed)

- $(1, 4, \pi) \in \mathbb{R}^3_+$
- $(0, -1, 5) \notin \mathbb{R}^3_+$ (negative is not allowed)

Also:

$$\mathbb{R}^n_{++} = \{\boldsymbol{x} \in \mathbb{R}^n : x_i > 0 \ \forall i\}.$$

This is strict positivity: no coordinate can be 0 ("exclude the boundary 0").

To talk about complements, we must first specify a universe set Ω: everything we discuss lies in Ω. In this course, commonly $\Omega = \mathbb{R}_+$ or $\Omega = \mathbb{R}$.

Definition 2.1.7. The complement of a set S is defined by

$$S^c = \{x \in \Omega : x \notin S\}.$$

Example 2.1.8. $S = (2, 3) \cup [5, 9)$.

- If $\Omega = \mathbb{R}$, then
$$S^c = (-\infty, 2] \cup [3, 5) \cup [9, \infty).$$
- If $\Omega = \mathbb{R}_+$, add the restriction $x \geq 0$:
$$S^c = [0, 2] \cup [3, 5) \cup [9, \infty).$$

We can define either open sets or closed sets first, and then define the other concept using complements. In this course, we use open balls in $\mathbb{R}^n$ as a working definition to build the notions of openness and closedness.

Definition 2.1.9 (Metric (Distance))**.** A metric $d(\boldsymbol{x}, \boldsymbol{y})$ measures the distance between points $\boldsymbol{x}$ and $\boldsymbol{y}$. Euclidean metric on $\mathbb{R}^n$:

$$d(\boldsymbol{x}, \boldsymbol{y}) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}.$$

Key defining properties (metric axioms):

- Triangle inequality:
$$d(\boldsymbol{x}, \boldsymbol{y}) \leq d(\boldsymbol{x}, \boldsymbol{z}) + d(\boldsymbol{z}, \boldsymbol{y}).$$
- Nonnegativity: $d(\boldsymbol{x}, \boldsymbol{y}) \geq 0$
- Identity of indiscernibles: $d(\boldsymbol{x}, \boldsymbol{y}) = 0 \iff \boldsymbol{x} = \boldsymbol{y}$
- Symmetry: $d(\boldsymbol{x}, \boldsymbol{y}) = d(\boldsymbol{y}, \boldsymbol{x})$

Example 2.1.10 (Illustration in $\mathbb{R}^2$)**.** If the coordinate differences between $\boldsymbol{x}$ and $\boldsymbol{y}$ are 1

horizontally and 2 vertically, then

$$d(\boldsymbol{x}, \boldsymbol{y}) = \sqrt{1^2 + 2^2} = \sqrt{5}.$$

Definition 2.1.11 (Open Balls)**.** Open ball around $\boldsymbol{x}$ with radius $\varepsilon > 0$:

$$B_\varepsilon(\boldsymbol{x}) = \{\boldsymbol{y} \in \mathbb{R}^n_+ : d(\boldsymbol{x}, \boldsymbol{y}) < \varepsilon\}.$$

This definition implies two important exclusions:

- It excludes the boundary (points with $d(\boldsymbol{x}, \boldsymbol{y}) = \varepsilon$ are not included because of "$<$").
- It also excludes points with negative coordinates, since the ball is defined with $\boldsymbol{y} \in \mathbb{R}^n_+$.

Example 2.1.12 (Using the Euclidean metric)**.** Distance between $(1, 0)$ and $(4, 5)$:

$$d\Big((1, 0), (4, 5)\Big) = \sqrt{(1-4)^2 + (0-5)^2} = \sqrt{9 + 25} = \sqrt{34}.$$

Open ball around $(2, 4)$ with $\varepsilon = 1$:

$$B_1(2, 4) = \{\, \boldsymbol{y} = (y_1, y_2) \in \mathbb{R}^2_+ : d\Big((2, 4), \boldsymbol{y}\Big) < 1 \,\}.$$

Equivalently (Euclidean distance),

$$B_1(2, 4) = \{\, (y_1, y_2) \in \mathbb{R}^2_+ : (y_1 - 2)^2 + (y_2 - 4)^2 < 1 \,\}.$$

Distance between $(1, 1, 1)$ and $(2, 4, 2)$:

$$d\Big((1, 1, 1), (2, 4, 2)\Big) = \sqrt{(1-2)^2 + (1-4)^2 + (1-2)^2} = \sqrt{1 + 9 + 1} = \sqrt{11}.$$

Definition 2.1.13 (Open Set)**.** $A \subseteq \mathbb{R}^n_+$ is open iff for every $\boldsymbol{x} \in A$ there is $\varepsilon_{\boldsymbol{x}} > 0$ such that

$$B_{\varepsilon_{\boldsymbol{x}}}(\boldsymbol{x}) \subseteq A.$$

Theorem 2.1.14. *Every open ball is open.*

Proof. Let $S = B_\varepsilon(\boldsymbol{x}^0)$. Take any $\boldsymbol{x} \in S$. Then $d(\boldsymbol{x}, \boldsymbol{x}^0) < \varepsilon$, so define

$$\delta = \varepsilon - d(\boldsymbol{x}, \boldsymbol{x}^0) > 0.$$

For any $\boldsymbol{y} \in B_\delta(\boldsymbol{x})$, by the triangle inequality,

$$d(\boldsymbol{y}, \boldsymbol{x}^0) \leq d(\boldsymbol{y}, \boldsymbol{x}) + d(\boldsymbol{x}, \boldsymbol{x}^0) < \delta + d(\boldsymbol{x}, \boldsymbol{x}^0) = \varepsilon,$$

hence $\boldsymbol{y} \in B_\varepsilon(\boldsymbol{x}^0) = S$. Therefore $B_\delta(\boldsymbol{x}) \subseteq S$, so S is open. $\square$

Definition 2.1.15 (Closed Set)**.** $A \subseteq \mathbb{R}^n_+$ is closed iff its complement is open.

Example 2.1.16. Complement of a small open ball in $\mathbb{R}^2_+$ is closed. In notation,

$$\Big(B_\varepsilon(\boldsymbol{x})\Big)^c = \{\, \boldsymbol{y} \in \mathbb{R}^2_+ : d(\boldsymbol{x}, \boldsymbol{y}) \geq \varepsilon \,\},$$

which is closed because it is the complement of an open set.

Example 2.1.17. Working in the universe $\mathbb{R}^n_+$:

- $\mathbb{R}^n_{++}$ is open in $\mathbb{R}^n_+$.
- $\varnothing$ is open in $\mathbb{R}^n_+$.
- Both are also closed (they are complements of each other):

$$(\mathbb{R}^n_{++})^c = \varnothing, \qquad \varnothing^c = \mathbb{R}^n_{++}.$$

The following questions highlight that "open/closed" depends on the chosen universe Ω (and complements are taken relative to Ω).

Example 2.1.18. Universe $[1, 2]$: is $[1, 2)$ open? closed?

Open in $[1, 2]$: YES. For any $x \in [1, 2)$, we can find $\varepsilon > 0$ such that $(x - \varepsilon, x + \varepsilon) \cap [1, 2] \subseteq [1, 2)$. (Equivalently, $[1, 2) = [1, 2] \cap (-\infty, 2)$, and $(-\infty, 2)$ is open in $\mathbb{R}$.)

Closed in $[1, 2]$: NO. The complement in the universe is $[1, 2] \setminus [1, 2) = \{2\}$, which is not open in $[1, 2]$. Hence $[1, 2)$ is not closed in $[1, 2]$.

Example 2.1.19. Universe $\mathbb{R}$: is $[1, 2)$ open? closed?

Open in $\mathbb{R}$: NO. (No open ball around 1 stays inside $[1, 2)$.)

Closed in $\mathbb{R}$: NO. (The point 2 is a limit point but is not included; equivalently, the complement is not open.)

Example 2.1.20. Open ball around the boundary point $\boldsymbol{x} = (0, 4)$ in $\mathbb{R}^2_+$:

In the universe $\mathbb{R}^2_+$, the open ball is the usual Euclidean ball intersected with $\mathbb{R}^2_+$:

$$B_\varepsilon(\boldsymbol{x}) = \{\boldsymbol{y} \in \mathbb{R}^2_+ : d(\boldsymbol{x}, \boldsymbol{y}) < \varepsilon\} = \{(y_1, y_2) \in \mathbb{R}^2_+ : y_1^2 + (y_2 - 4)^2 < \varepsilon^2\}.$$

Geometrically, it looks like a "half-disk" (the part with $y_1 \geq 0$) because points with negative y_1 are excluded by the universe $\mathbb{R}^2_+$.

Theorem 2.1.21. *$\varnothing$ and $\mathbb{R}^n$ are open.*

Proof. $\mathbb{R}^n$: for any $\boldsymbol{x} \in \mathbb{R}^n$, any ball $B_\varepsilon(\boldsymbol{x}) \subseteq \mathbb{R}^n$. $\varnothing$: vacuously open (there is no $\boldsymbol{x} \in \varnothing$ to check). $\square$

Theorem 2.1.22. *Union of any family of open sets is open.*

Proof. If $\boldsymbol{x} \in \bigcup_\alpha A_\alpha$, then $\boldsymbol{x} \in A_{\alpha_0}$ for some α_0. Since A_{α_0} is open, $\exists \varepsilon > 0$ with $B_\varepsilon(\boldsymbol{x}) \subseteq A_{\alpha_0} \subseteq \bigcup_\alpha A_\alpha$. □

Theorem 2.1.23. *Intersection of finitely many open sets is open.*

Proof. If $\boldsymbol{x} \in \bigcap_{i=1}^m A_i$, then for each i there exists $\varepsilon_i > 0$ with $B_{\varepsilon_i}(\boldsymbol{x}) \subseteq A_i$. Let $\varepsilon = \min\{\varepsilon_1, \ldots, \varepsilon_m\}$. Then $B_\varepsilon(\boldsymbol{x}) \subseteq A_i$ for all i, hence $B_\varepsilon(\boldsymbol{x}) \subseteq \bigcap_{i=1}^m A_i$. □

Example 2.1.24 (why "finite" matters)**.** Consider $U_k = (-1/k, 1/k)$. Let $U_k = (-1/k, 1/k)$, each U_k is open in $\mathbb{R}$. But

$$\bigcap_{k=1}^{\infty} U_k = \{0\},$$

and $\{0\}$ is not open in $\mathbb{R}$.

This shows: an infinite intersection of open sets need not be open, which is why the slide states "intersection of finitely many open sets is open."

To relate unions/intersections with complements, we use De Morgan's laws:

Definition 2.1.25 (De Morgan's Laws)**.**

$$\left(\bigcup_i A_i\right)^c = \bigcap_i A_i^c, \qquad \left(\bigcap_i A_i\right)^c = \bigcup_i A_i^c.$$

These identities help transfer "open-set" properties to "closed-set" properties via complements. In particular:

- Closed sets: $\varnothing$, $\mathbb{R}^n$ are closed.
- Finite unions of closed sets are closed.
- Arbitrary intersections of closed sets are closed.

Definition 2.1.26. A set A is bounded if it is contained inside some open ball.

Theorem 2.1.27. *In Euclidean spaces (like $\mathbb{R}^n_+$), a set is compact iff it is closed and bounded.*

Definition 2.1.28 (ε-δ Continuity)**.** Let $f : D \to \mathbb{R}$. The function f is continuous at $\boldsymbol{x}$ iff for all $\varepsilon > 0$ there exists $\delta > 0$ s.t.

$$d(\boldsymbol{x}, \boldsymbol{y}) < \delta \;\Rightarrow\; |f(\boldsymbol{x}) - f(\boldsymbol{y})| < \varepsilon.$$

Definition 2.1.29 (Continuity Equivalent Definition)**.** To connect continuity with open sets, let $D \subseteq \mathbb{R}^n_+$. A function $f : D \to \mathbb{R}$ is continuous iff the inverse image

$$f^{-1}(B) = \{\boldsymbol{x} \in D : f(\boldsymbol{x}) \in B\}$$

of each open ball B in the range $\mathbb{R}$ is also open in the domain D.

Definition 2.1.30 (Convex Set)**.** A set $A \subseteq \mathbb{R}^n_+$ is convex iff for all $\boldsymbol{x}, \boldsymbol{y} \in A$ and $\alpha \in [0, 1]$,

$$\alpha\boldsymbol{x} + (1 - \alpha)\boldsymbol{y} \in A.$$

2.2 Preferences

Let the consumption set be $\mathbb{R}^n_+$. A preference relation is a subset

$$\succeq \subseteq \mathbb{R}^n_+ \times \mathbb{R}^n_+.$$

$(\boldsymbol{x}, \boldsymbol{y}) \in \succeq$ is written as $\boldsymbol{x} \succeq \boldsymbol{y}$. Meaning: bundle $\boldsymbol{x}$ is at least as good as bundle $\boldsymbol{y}$.

Example 2.2.1. Four canonical examples:

1. $\succeq_1 = \mathbb{R}^n_+ \times \mathbb{R}^n_+$: everything is at least as good as everything.
2. $\succeq_2 = \varnothing$: nothing is comparable (even to itself).
3. $\succeq_3 = \{(\boldsymbol{x}, \boldsymbol{y}) : x_i \geq y_i \ \forall i\}$: coordinate-wise order.
4. $\succeq_4 = \{(\boldsymbol{x}, \boldsymbol{y}) : \prod_{i=1}^n x_i \geq \prod_{i=1}^n y_i\}$: product index order.

In $\mathbb{R}^2_+$, let $\boldsymbol{x} = (1, 2)$ and $\boldsymbol{y} = (2, 1)$.

- Under $\succeq_1$: $\boldsymbol{x} \succeq \boldsymbol{y}$ and $\boldsymbol{y} \succeq \boldsymbol{x}$.
- Under $\succeq_3$: $\boldsymbol{x}$ and $\boldsymbol{y}$ are incomparable.
- Under $\succeq_4$: compare products:

$$(1)(2) = 2, \qquad (2)(1) = 2 \Rightarrow \boldsymbol{x} \succeq_4 \boldsymbol{y} \text{ and } \boldsymbol{y} \succeq_4 \boldsymbol{x}$$

  (i.e., they are tied / indifferent under the product index order).

We define "best" using a weak relation:

Definition 2.2.2. $\boldsymbol{x} \succeq \boldsymbol{y}$: $\boldsymbol{x}$ is at least as good as $\boldsymbol{y}$.

This choice is convenient because: It captures both strict rankings and ties. Verbal interpretation: "better", "worse", or "about the same".

Starting from the weak preference relation $\succeq$, we define two derived relations.

Definition 2.2.3. Strict preference:

$$\boldsymbol{x} \succ \boldsymbol{y} \iff \boldsymbol{x} \succeq \boldsymbol{y} \text{ and not}(\boldsymbol{y} \succeq \boldsymbol{x}).$$

Indifference:

$$\boldsymbol{x} \sim \boldsymbol{y} \iff \boldsymbol{x} \succeq \boldsymbol{y} \text{ and } \boldsymbol{y} \succeq \boldsymbol{x}.$$

(Both directions hold $\Rightarrow \boldsymbol{x} \sim \boldsymbol{y}$.)

What Happens in the Four Examples? Applying these definitions to the four canonical relations:

Relation	Indifference $\sim$	Strict preference $\succ$
$\succeq_1$	all pairs	empty
$\succeq_2$	empty	empty
$\succeq_3$	only identical bundles	coordinate-wise $\geq$ with at least one strict
$\succeq_4$	equal products	higher product

Definition 2.2.4 (Lexicographic Preferences)**.** In $\mathbb{R}^2_+$, define lexicographic weak preference by:

$$\boldsymbol{x} \succeq_L \boldsymbol{y} \iff (x_1 > y_1) \text{ or } (x_1 = y_1 \text{ and } x_2 \geq y_2).$$

This means: compare the first component first; use the second component only when the first components tie.

Example 2.2.5. 1. $(1, 2)$ vs $(2, 1)$: the second bundle is strictly preferred (larger first component).

2. $(1, 2)$ vs $(1, 1)$: first components tie, so compare second; $(1, 2)$ is preferred.

3. $(1, 2)$ vs $(1, 2)$: indifferent.

From $\succeq_L$, the derived relations are:

- Indifference:
$$\boldsymbol{x} \sim_L \boldsymbol{y} \iff \boldsymbol{x} = \boldsymbol{y}$$
(only identical bundles are indifferent).
- Strict preference:
$$\boldsymbol{x} \succ_L \boldsymbol{y} \iff x_1 > y_1 \text{ or } (x_1 = y_1 \text{ and } x_2 > y_2).$$
- Geometrically, "indifference curves" collapse to single points.

There are as many possible preferences as there are subsets of $\mathbb{R}^n_+ \times \mathbb{R}^n_+$. We restrict attention to "reasonable" preferences that satisfy some axioms.

Axiom 2.2.6 (Complete preferences)**.** A preference relation $\succeq$ on $\mathbb{R}^n_+$ is complete if:

$$\forall \boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n_+, \quad \boldsymbol{x} \succeq \boldsymbol{y} \text{ or } \boldsymbol{y} \succeq \boldsymbol{x} \text{ (or both)}.$$

Interpretation: preferences are rich enough to compare any pair. Economic question: is this always behaviorally realistic?

Example 2.2.7 (Which Examples Satisfy It?)**.** • $\succeq_1$: complete (both directions always hold).

- $\succeq_2$: not complete (no comparisons hold).
- $\succeq_3$: not complete (e.g., $(1, 2)$ and $(2, 1)$).
- $\succeq_4$: complete (compare the real numbers $\prod_i x_i$ and $\prod_i y_i$).
- $\succeq_L$: complete.

Axiom 2.2.8 (Transitive preferences)**.** A preference relation $\succeq$ on $\mathbb{R}^n_+$ is transitive if:

$$\forall \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathbb{R}^n_+, \quad (\boldsymbol{x} \succeq \boldsymbol{y} \text{ and } \boldsymbol{y} \succeq \boldsymbol{z}) \Rightarrow \boldsymbol{x} \succeq \boldsymbol{z}.$$

Interpretation: no preference cycles in ranking logic.

Example 2.2.9. • $\succeq_1$: transitive.

- $\succeq_2$: transitive (vacuously: antecedent never true).
- $\succeq_3$: transitive (coordinate-wise inequalities chain).
- $\succeq_4$: transitive (real-number order of product index).
- $\succeq_L$: transitive.

Theorem 2.2.10 (Transitivity Transfers)**.** *If $\succeq$ is transitive, then:*

1. *Indifference $\sim$ is transitive.*
2. *Strict preference $\succ$ is transitive.*
3. *If $\boldsymbol{x} \succeq \boldsymbol{y}$ and $\boldsymbol{y} \succeq \boldsymbol{z}$ with at least one strict, then $\boldsymbol{x} \succ \boldsymbol{z}$.*

Proof. (1) $\sim$ is transitive

Recall: $\boldsymbol{x} \sim \boldsymbol{y} \iff (\boldsymbol{x} \succeq \boldsymbol{y} \text{ and } \boldsymbol{y} \succeq \boldsymbol{x})$. Assume $\boldsymbol{x} \sim \boldsymbol{y}$ and $\boldsymbol{y} \sim \boldsymbol{z}$. Then

- $\boldsymbol{x} \succeq \boldsymbol{y}$ and $\boldsymbol{y} \succeq \boldsymbol{x}$,
- $\boldsymbol{y} \succeq \boldsymbol{z}$ and $\boldsymbol{z} \succeq \boldsymbol{y}$.

By transitivity of $\succeq$:

- from $\boldsymbol{x} \succeq \boldsymbol{y}$ and $\boldsymbol{y} \succeq \boldsymbol{z}$, we get $\boldsymbol{x} \succeq \boldsymbol{z}$;
- from $\boldsymbol{z} \succeq \boldsymbol{y}$ and $\boldsymbol{y} \succeq \boldsymbol{x}$, we get $\boldsymbol{z} \succeq \boldsymbol{x}$.

Hence $\boldsymbol{x} \sim \boldsymbol{z}$. So $\sim$ is transitive.

(2) $\succ$ is transitive (strict transitivity)

Recall: $\boldsymbol{x} \succ \boldsymbol{y} \iff (\boldsymbol{x} \succeq \boldsymbol{y} \text{ and } \neg(\boldsymbol{y} \succeq \boldsymbol{x}))$. To show: $\boldsymbol{x} \succ \boldsymbol{y}$ and $\boldsymbol{y} \succ \boldsymbol{z} \Rightarrow \boldsymbol{x} \succ \boldsymbol{z}$. Assume $\boldsymbol{x} \succ \boldsymbol{y}$ and $\boldsymbol{y} \succ \boldsymbol{z}$. Then:

- $\boldsymbol{x} \succeq \boldsymbol{y}$ and $\neg(\boldsymbol{y} \succeq \boldsymbol{x})$,
- $\boldsymbol{y} \succeq \boldsymbol{z}$ and $\neg(\boldsymbol{z} \succeq \boldsymbol{y})$.

By transitivity of $\succeq$, from $\boldsymbol{x} \succeq \boldsymbol{y}$ and $\boldsymbol{y} \succeq \boldsymbol{z}$, we get $\boldsymbol{x} \succeq \boldsymbol{z}$.

Now show $\neg(\boldsymbol{z} \succeq \boldsymbol{x})$ by contradiction. If $\boldsymbol{z} \succeq \boldsymbol{x}$, then with $\boldsymbol{x} \succeq \boldsymbol{y}$, transitivity gives $\boldsymbol{z} \succeq \boldsymbol{y}$, contradicting $\neg(\boldsymbol{z} \succeq \boldsymbol{y})$. Therefore $\neg(\boldsymbol{z} \succeq \boldsymbol{x})$, so $\boldsymbol{x} \succ \boldsymbol{z}$.

(3) If $\boldsymbol{x} \succeq \boldsymbol{y}$ and $\boldsymbol{y} \succeq \boldsymbol{z}$ with at least one strict, then $\boldsymbol{x} \succ \boldsymbol{z}$

Assume $\boldsymbol{x} \succeq \boldsymbol{y}$ and $\boldsymbol{y} \succeq \boldsymbol{z}$. By transitivity, $\boldsymbol{x} \succeq \boldsymbol{z}$. It remains to prove $\neg(\boldsymbol{z} \succeq \boldsymbol{x})$. Consider the two possible "strict" cases:

- Case A: $\boldsymbol{x} \succ \boldsymbol{y}$ (so $\neg(\boldsymbol{y} \succeq \boldsymbol{x})$). Suppose $\boldsymbol{z} \succeq \boldsymbol{x}$. Since $\boldsymbol{y} \succeq \boldsymbol{z}$, transitivity implies $\boldsymbol{y} \succeq \boldsymbol{x}$, contradicting $\neg(\boldsymbol{y} \succeq \boldsymbol{x})$. Hence $\neg(\boldsymbol{z} \succeq \boldsymbol{x})$, so $\boldsymbol{x} \succ \boldsymbol{z}$.
- Case B: $\boldsymbol{y} \succ \boldsymbol{z}$ (so $\neg(\boldsymbol{z} \succeq \boldsymbol{y})$). Suppose $\boldsymbol{z} \succeq \boldsymbol{x}$. Since $\boldsymbol{x} \succeq \boldsymbol{y}$, transitivity implies $\boldsymbol{z} \succeq \boldsymbol{y}$, contradicting $\neg(\boldsymbol{z} \succeq \boldsymbol{y})$. Hence $\neg(\boldsymbol{z} \succeq \boldsymbol{x})$, so $\boldsymbol{x} \succ \boldsymbol{z}$.

Therefore, in either case, $\boldsymbol{x} \succ \boldsymbol{z}$. □

Axioms 2.2.6 and 2.2.8 are enough to have a utility representation if Ω is countable! But for $\mathbb{R}^n_+$ we need:

Axiom 2.2.11 (Continuity)**.** A preference relation $\succeq$ on $\mathbb{R}^n_+$ is continuous iff for each $\boldsymbol{x} \in \mathbb{R}^n_+$, the sets

$$G(\boldsymbol{x}) = \{\boldsymbol{y} \in \mathbb{R}^n_+ : \boldsymbol{y} \succeq \boldsymbol{x}\}, \qquad W(\boldsymbol{x}) = \{\boldsymbol{y} \in \mathbb{R}^n_+ : \boldsymbol{x} \succeq \boldsymbol{y}\}$$

are closed in $\mathbb{R}^n_+$.

- $G(\boldsymbol{x})$: "at least as good as $\boldsymbol{x}$" set.
- $W(\boldsymbol{x})$: "no better than $\boldsymbol{x}$" set.

Geometric Meaning of Continuity: Think of an indifference curve through $\boldsymbol{x}$.

- $G(\boldsymbol{x})$: region on and above the curve.
- $W(\boldsymbol{x})$: region on and below the curve.

Continuity asks both regions to be closed.

Example 2.2.12. Classify these relations:

1. $\boldsymbol{x} \succeq \boldsymbol{y} \iff x_1 x_2 \geq y_1 y_2$.
2. $\boldsymbol{x} \succeq \boldsymbol{y} \iff x_1^2 + x_2 \geq y_1^2 + y_2$.
3. Lexicographic $\succeq_L$.
4. $\boldsymbol{x} \succeq \boldsymbol{y} \iff (x_1 - 3)^2 + (x_2 - 4)^2 \geq (y_1 - 3)^2 + (y_2 - 4)^2$.

Which Are Complete, Transitive, Continuous? (on $\mathbb{R}^2_+$)

- (1) Complete: yes. Transitive: yes. Continuous: yes.
- (2) Complete: yes. Transitive: yes. Continuous: yes.

- (3) Complete: yes. Transitive: yes. Continuous: no.
- (4) Complete: yes. Transitive: yes. Continuous: yes.

Theorem 2.2.13 (Open Strict Sets Characterization)**.** *Suppose preferences are complete. Then continuity is equivalent to:*

$$SG(\boldsymbol{x}) = \{\boldsymbol{y} \in \mathbb{R}^n_+ : \boldsymbol{y} \succ \boldsymbol{x}\} \text{ is open}, \qquad SW(\boldsymbol{x}) = \{\boldsymbol{y} \in \mathbb{R}^n_+ : \boldsymbol{x} \succ \boldsymbol{y}\} \text{ is open}.$$

Why? Under completeness:

$$SG(\boldsymbol{x}) = W(\boldsymbol{x})^c, \qquad SW(\boldsymbol{x}) = G(\boldsymbol{x})^c.$$

Theorem 2.2.14 (Intermediate-Value Version)**.** *If preferences are complete and continuous, then:*

$$\boldsymbol{x} \succ \boldsymbol{y} \succ \boldsymbol{z} \;\Rightarrow\; \exists \alpha \in (0,1) \text{ such that } \alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{z} \sim \boldsymbol{y}.$$

Proof. Define

$$A = \{\alpha \in [0,1] : \boldsymbol{y} \succ \alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{z}\}.$$

- $0 \in A$ because $\boldsymbol{y} \succ \boldsymbol{z}$ and $\alpha = 0$ implies $\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{z} = \boldsymbol{z}$.
- $A \subseteq [0,1]$, so A is bounded above by 1.
- By the least upper bound property, the supremum exists:

$$\alpha^* = \sup A.$$

Goal: prove $0 < \alpha^* < 1$.

- Show $\alpha^* > 0$:

  Since $\boldsymbol{y} \succ \boldsymbol{z}$, we have $\boldsymbol{z} \in SW(\boldsymbol{y})$, where

  $$SW(\boldsymbol{y}) = \{\boldsymbol{w} : \boldsymbol{y} \succ \boldsymbol{w}\}.$$

  Under completeness + continuity (Result 2), strict sets like $SW(\boldsymbol{y})$ are open. Hence there exists a neighborhood of $\boldsymbol{z}$ (an open ball around $\boldsymbol{z}$) contained in $SW(\boldsymbol{y})$. Points on the segment $\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{z}$ with $\alpha > 0$ sufficiently small are close to $\boldsymbol{z}$, so they also lie in $SW(\boldsymbol{y})$. Therefore, for some small $\bar{\alpha} > 0$,

  $$\boldsymbol{y} \succ \bar{\alpha}\boldsymbol{x} + (1-\bar{\alpha})\boldsymbol{z},$$

  i.e., $\bar{\alpha} \in A$. Hence A contains a positive number, so $\alpha^* = \sup A > 0$.

- Show $\alpha^* < 1$:

  Since $\boldsymbol{x} \succ \boldsymbol{y}$, we have $\boldsymbol{x} \in SG(\boldsymbol{y})$, where

  $$SG(\boldsymbol{y}) = \{\boldsymbol{w} : \boldsymbol{w} \succ \boldsymbol{y}\}.$$

  Again, under completeness + continuity, $SG(\boldsymbol{y})$ is open, so there is a neighborhood of $\boldsymbol{x}$ contained in $SG(\boldsymbol{y})$. Points on the segment with α sufficiently close to 1 are close to $\boldsymbol{x}$, hence lie in $SG(\boldsymbol{y})$,

meaning they satisfy $\boldsymbol{w} \succ \boldsymbol{y}$ and therefore cannot satisfy $\boldsymbol{y} \succ \boldsymbol{w}$. Thus there exists $\eta > 0$ such that for all $\alpha \in (1-\eta, 1]$,

$$\alpha \notin A.$$

So A is bounded above by $1-\eta < 1$, implying $\alpha^* < 1$.

Define

$$\boldsymbol{c}^* = \alpha^* \boldsymbol{x} + (1-\alpha^*)\boldsymbol{z}.$$

By completeness, exactly one of the following must hold:

$$\boldsymbol{c}^* \succ \boldsymbol{y}, \qquad \boldsymbol{y} \succ \boldsymbol{c}^*, \qquad \boldsymbol{c}^* \sim \boldsymbol{y}.$$

We rule out the first two using continuity (openness of strict sets) plus the supremum property.

- Rule out $\boldsymbol{y} \succ \boldsymbol{c}^*$:

  If $\boldsymbol{y} \succ \boldsymbol{c}^*$, then $\boldsymbol{c}^* \in SW(\boldsymbol{y})$. Since $SW(\boldsymbol{y})$ is open, there exists a neighborhood of $\boldsymbol{c}^*$ contained in $SW(\boldsymbol{y})$. Points on the segment slightly past $\boldsymbol{c}^*$ toward $\boldsymbol{x}$ (i.e., with $\alpha > \alpha^*$ but very close) would also lie in $SW(\boldsymbol{y})$, implying those $\alpha > \alpha^*$ belong to A. That contradicts α^* being an upper bound (indeed the least upper bound) of A. Hence $\boldsymbol{y} \not\succ \boldsymbol{c}^*$.

- Rule out $\boldsymbol{c}^* \succ \boldsymbol{y}$:

  If $\boldsymbol{c}^* \succ \boldsymbol{y}$, then $\boldsymbol{c}^* \in SG(\boldsymbol{y})$. Since $SG(\boldsymbol{y})$ is open, there exists a neighborhood of $\boldsymbol{c}^*$ contained in $SG(\boldsymbol{y})$. In particular, along the segment near $\boldsymbol{c}^*$ from the $\boldsymbol{z}$-side, there is an interval $(\alpha^* - \delta, \alpha^*)$ such that for all α in that interval,

  $$\alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{z} \in SG(\boldsymbol{y}) \Rightarrow \neg\Big(\boldsymbol{y} \succ \alpha\boldsymbol{x} + (1-\alpha)\boldsymbol{z}\Big),$$

  so no $\alpha \in (\alpha^* - \delta, \alpha^*)$ can lie in A. But this contradicts the definition of $\alpha^* = \sup A$, which guarantees that A has elements arbitrarily close to α^* from below. Hence $\boldsymbol{c}^* \not\succ \boldsymbol{y}$.

Therefore the only remaining case is:

$$\boldsymbol{c}^* \sim \boldsymbol{y}.$$

□

This is a strong statement: an indifference point appears on the segment from $\boldsymbol{z}$ to $\boldsymbol{x}$. This motivates continuity as a no-jumps assumption.

Why "Intermediate Value"? It compares with the Intermediate Value Theorem (IVT) for continuous real functions. If u represents preferences and is continuous, then

$$\boldsymbol{x} \succ \boldsymbol{y} \succ \boldsymbol{z} \Rightarrow u(\boldsymbol{x}) > u(\boldsymbol{y}) > u(\boldsymbol{z}).$$

IVT then gives a point on the segment $[\boldsymbol{z}, \boldsymbol{x}]$ with utility $u(\boldsymbol{y})$. (Graphically: utility as a function of the segment parameter crosses the level $u(\boldsymbol{y})$.)

2.3 Utility

Theorem 2.3.1. *If preferences can be represented with a continuous utility function, then those preferences satisfy Axiom 2.2.11 (continuity).*

Remark. It can help us determine whether preferences are continuous.

Proof. Let $u : \mathbb{R}^n_+ \to \mathbb{R}$ represent $\succeq$, i.e.

$$\boldsymbol{x} \succeq \boldsymbol{y} \iff u(\boldsymbol{x}) \geq u(\boldsymbol{y}).$$

Fix $\boldsymbol{x}$. Then the upper and lower contour sets become:

$$G(\boldsymbol{x}) = \{\boldsymbol{y} : \boldsymbol{y} \succeq \boldsymbol{x}\} = \{\boldsymbol{y} : u(\boldsymbol{y}) \geq u(\boldsymbol{x})\} = u^{-1}([u(\boldsymbol{x}), \infty)),$$

$$W(\boldsymbol{x}) = \{\boldsymbol{y} : \boldsymbol{x} \succeq \boldsymbol{y}\} = \{\boldsymbol{y} : u(\boldsymbol{y}) \leq u(\boldsymbol{x})\} = u^{-1}((-\infty, u(\boldsymbol{x})]).$$

The sets $[u(\boldsymbol{x}), \infty)$ and $(-\infty, u(\boldsymbol{x})]$ are closed in $\mathbb{R}$. If u is continuous, the preimage of a closed set is closed. Hence $G(\boldsymbol{x})$ and $W(\boldsymbol{x})$ are closed for every $\boldsymbol{x}$, which is exactly Axiom 2.2.11 (closed-set form). □

Theorem 2.3.2. *If a preference relation $\succeq$ is complete, then $u : \mathbb{R}^n_+ \to \mathbb{R}$ is a utility representation of $\succeq$ iff for all bundles $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n_+$:*

1. *If $\boldsymbol{x} \succ \boldsymbol{y}$ then $u(\boldsymbol{x}) > u(\boldsymbol{y})$.*
2. *If $u(\boldsymbol{x}) > u(\boldsymbol{y})$ then $\boldsymbol{x} \succ \boldsymbol{y}$.*

Is completeness needed? Yes!

Axiom 2.3.3 (Strong Monotonicity)**.** If $x_i \geq y_i$ for all i and $x_j > y_j$ for some j, then $\boldsymbol{x} \succ \boldsymbol{y}$.

In short, "more of at least one good, no less of others" implies strictly better. It is stronger than weak monotonicity formulations.

Theorem 2.3.4 (U-Representation)**.** *If preferences are:*

- *complete,*
- *transitive, and*
- *continuous,*

then a utility representation exists on $\mathbb{R}^n_+$.

Remark. Finite set is not needed!

Proof. If we add in strong monotonicity the proof is a little simpler (see supplementary notes). □

Note: The proof is complicated, but you should be able to show the reverse direction (Continuous U exists $\Rightarrow$ Axioms 1–3 hold).

Proof. Reverse direction: If u represents $\succeq$ in the standard sense $\boldsymbol{x} \succeq \boldsymbol{y} \iff u(\boldsymbol{x}) \geq u(\boldsymbol{y})$, then:

- Completeness: for any $\boldsymbol{x}, \boldsymbol{y}$, either $u(\boldsymbol{x}) \geq u(\boldsymbol{y})$ or $u(\boldsymbol{y}) \geq u(\boldsymbol{x})$; hence $\boldsymbol{x} \succeq \boldsymbol{y}$ or $\boldsymbol{y} \succeq \boldsymbol{x}$.
- Transitivity: if $\boldsymbol{x} \succeq \boldsymbol{y}$ and $\boldsymbol{y} \succeq \boldsymbol{z}$, then $u(\boldsymbol{x}) \geq u(\boldsymbol{y}) \geq u(\boldsymbol{z})$, so $u(\boldsymbol{x}) \geq u(\boldsymbol{z})$, hence $\boldsymbol{x} \succeq \boldsymbol{z}$.
- Continuity (Axiom 3): if u is continuous, then for each $\boldsymbol{x}$,
$$G(\boldsymbol{x}) = \{\boldsymbol{y} : \boldsymbol{y} \succeq \boldsymbol{x}\} = \{\boldsymbol{y} : u(\boldsymbol{y}) \geq u(\boldsymbol{x})\} = u^{-1}([u(\boldsymbol{x}), \infty))$$
and
$$W(\boldsymbol{x}) = \{\boldsymbol{y} : \boldsymbol{x} \succeq \boldsymbol{y}\} = \{\boldsymbol{y} : u(\boldsymbol{y}) \leq u(\boldsymbol{x})\} = u^{-1}((-\infty, u(\boldsymbol{x})])$$
are preimages of closed sets, hence closed.

□

On finite/countable sets: complete + transitive already suffice. If represented by a utility function, preferences are complete and transitive. If represented by a continuous utility, preferences are continuous.

Utility representation is generally not unique. Any strictly increasing transformation preserves preferences. Different representations can simplify different tasks.

Example 2.3.5. Logs can linearize multiplicative forms (same preferences: u vs $\log u$).

In optimization, smooth forms make derivatives and FOCs cleaner.

3 Quasi-concavity and U-max

3.1 Setup

Definition 3.1.1 (Consumer Problem)**.** Find $x^* \in B(p, I) \subseteq \mathbb{R}^n_+$ such that
$$x^* \succeq y \quad \text{for all } y \in B(p, I).$$

This is a strong optimality notion: x^* is at least as good as every feasible bundle. Alternative (weaker) notion: no feasible bundle is strictly better. In this course we use the strong notion under complete preferences.

Definition 3.1.2. Given prices $p = (p_1, \ldots, p_n) \in \mathbb{R}^n_{++}$ and income $I \in \mathbb{R}_+$:

- Budget set:
$$B(p, I) = \{x \in \mathbb{R}^n_+ : p \cdot x \leq I\}.$$
- Budget line:
$$p \cdot x = I.$$

For 2-good, the buget line is $p_1 x_1 + p_2 x_2 = I$. In n goods, this is a hyperplane boundary of $B(p, I)$.

To guarantee a solution to utility maximization,
$$\max\{u(x) : x \in B(p, I)\},$$
we use:

Theorem 3.1.3 (Weierstrass Theorem)**.** *A continuous real-valued function on a non-empty compact set attains a maximum (and minimum).*

So we need: continuity of u, and non-empty compact $B(p, I)$.

To support existence results (e.g., via compactness of the feasible set), we verify that the budget set $B(p, \ell)$ is compact when prices are strictly positive.

1. Non-empty: $0 \in B(p, \ell)$ whenever $\ell \geq 0$.
2. Closed: $\{x : p \cdot x \leq \ell\}$ is closed; its intersection with $\mathbb{R}^n_+$ is also closed.
3. Bounded: with $p_i > 0$, each coordinate satisfies

$$x_i \leq \frac{\ell}{p_i}.$$

Hence $B(p, \ell)$ is (with strictly positive prices) a compact triangle-like region (in the 2-good picture).

Recall that a set $A \subseteq \mathbb{R}^n$ is convex if every line segment between two points in A stays in A. Formally, $A \subseteq \mathbb{R}^n$ is convex iff for all $x, y \in A$ and $\alpha \in [0, 1]$,

$$\alpha x + (1 - \alpha)y \in A.$$

Example 3.1.4.
- Convex: balls, boxes, half-spaces, budget sets with linear prices.
- Non-convex: donut/ring shapes (with a "hole"), kinked discount sets.

Theorem 3.1.5. *$B(p, I)$ is convex.*

Proof. If $x, y \in B(p, I)$ and $t \in [0, 1]$, then

$$p \cdot (tx + (1 - t)y) = t\, p \cdot x + (1 - t)\, p \cdot y \leq tI + (1 - t)I = I.$$

So $tx + (1 - t)y \in B(p, I)$. □

Non-convex budget sets can occur. Convexity can fail when prices depend on quantity.

Example 3.1.6. Bulk discounts after thresholds. The set may remain closed and bounded, but develop a "dent" (hence become non-convex).

Existence can still hold; however, simple FOC methods may fail.

Theorem 3.1.7 (Weierstrass's Theorem)**.** *Every continuous real-valued function f on a non-empty compact set $B \subset \mathbb{R}^n$ attains a maximum value on B. That is, there exists some $x \in B$ such that*

$$f(x) \geq f(y) \quad \forall y \in B.$$

This is an existence result, not yet uniqueness. Multiple maximizers are possible without strict curvature assumptions.

Theorem 3.1.8 (Weierstrass's theorem (consumer choice version))**.** *If preferences admit a continuous utility representation u, and the budget set $B(p, I)$ is non-empty and compact, then an optimal bundle exists.*

Below checks connect directly to the assumptions behind Weierstrass (continuity + non-empty compact feasible set).

1. Is the budget line compact in $\mathbb{R}^n$?

   No. In $\mathbb{R}^n$, the budget line $\{x \in \mathbb{R}^n : p \cdot x = I\}$ is a hyperplane, which is closed but unbounded, hence not compact.

2. Can existence fail if the feasible set is open (not closed)?

   Yes. A maximum may not be attained on an open set (e.g., maximizing a continuous objective can yield only a supremum on the boundary that is not included).

3. Can existence fail if the set is unbounded?

   Yes. If the feasible set is unbounded and utility can increase without bound along feasible directions, there may be no maximizer (only "larger and larger" feasible values).

4. Can existence fail if utility is discontinuous?

   Yes. Even on a compact feasible set, discontinuity can prevent attainment of a maximum (e.g., the function may approach arbitrarily high values near a point without achieving them).

3.2 Convex Preferences and Quasi-Convavity

Building on the notion of convex sets, we define convexity for preferences via upper contour sets.

Definition 3.2.1. Preferences $\succeq$ are convex if for every $x \in \mathbb{R}^n_+$, the upper contour set

$$G(x) = \{y \in \mathbb{R}^n_+ : y \succeq x\}$$

is convex.

Economic interpretation: averages are at least as good as extremes.

Remark. For complete and transitive preferences, convexity is equivalent to:

$$x \succeq y \Rightarrow tx + (1-t)y \succeq y, \qquad \forall t \in [0, 1].$$

In words, if x is at least as good as y, then every average with y is also at least as good as y.

Strengthening convexity leads to strict convexity.

Definition 3.2.2. $\succeq$ is strictly convex if for all distinct x, y and all $t \in (0, 1)$:

$$x \succeq y \Rightarrow tx + (1-t)y \succ y.$$

Economic interpretation: averages are strictly better than extremes.

This diagram illustrates the difference between convex (weak) and strictly convex behavior using two bundles x and y in (x_1, x_2)-space.

- Convex (weak): the line segment between x and y is drawn as a straight segment, indicating that taking averages (mixtures) of two bundles need only be no worse than the extremes; the "average" can lie on the same flat segment (so not strictly better).
- Strictly convex: the curve is bowed, and the mixture point $tx + (1-t)y$ is shown strictly above the curve through x and y, indicating that averages are strictly better than extremes (for $t \in (0,1)$).

Remark. With diminishing MRS and "more is better", we can say the preference is convex, but not the other way around.

We now move to Quasi-Concavity.

Careful: convexity of a function is not the same as convexity of preferences.

Definition 3.2.3 (Function concavity)**.** A real-valued function is concave if for any x, y on its domain,

$$f(tx + (1-t)y) \geq tf(x) + (1-t)f(y) \qquad \forall t \in [0,1].$$

The region "below" the graph is convex for a concave function; the set "above" is convex for a convex function.

Definition 3.2.4 (Quasi-Concavity of Utility)**.** $u : D \to \mathbb{R}$, where D is a convex subset of $\mathbb{R}^n_+$ is quasi-concave if for all x, y and $t \in [0,1]$,

$$u(x) \geq u(y) \Rightarrow u(tx + (1-t)y) \geq u(y).$$

Remark. Equivalent: all upper contour sets $\{z : u(z) \geq c\}$ are convex.

Definition 3.2.5 (Strict Quasi-Concavity)**.** u is strictly quasi-concave if for all distinct x, y and all $t \in (0,1)$,

$$u(x) \geq u(y) \Rightarrow u(tx + (1-t)y) > u(y).$$

Implication: guarantees uniqueness of the maximizer on convex feasible sets.

The relationship between concavity and quasi-concavity:

- Every concave function is quasi-concave.
- Every strictly concave function is strictly quasi-concave.
- The converse is false: quasi-concavity is weaker.

Example 3.2.6. $u(x) = x^2$ on $\mathbb{R}_+$.

On $\mathbb{R}_+$, x^2 is strictly increasing. **Any strictly increasing 1D function is quasi-concave on its domain** (upper contour sets are rays/intervals). Therefore $u(x) = x^2$ is quasi-concave (indeed strictly quasi-concave) on $\mathbb{R}_+$, despite being convex as a function.

Example 3.2.7. $u(x_1, x_2) = x_1^2 + x_2$. Is it concave? No.

Is it quasi-concave on $\mathbb{R}^2_+$? No. (Even though both x_1^2 and x_2 are quasi-concave.)

Upper contour sets: for any c,

$$\{(x_1, x_2) \in \mathbb{R}^2_+ : x_1^2 + x_2 \geq c\} = \{(x_1, x_2) \in \mathbb{R}^2_+ : x_2 \geq c - x_1^2\}.$$

This is the region above the curve $x_2 = c - x_1^2$ (restricted to $x_1 \geq 0$, $x_2 \geq 0$).

Direct non-convexity check (counterexample): take $c = 1$.

- $A = (0, 1)$ satisfies $0^2 + 1 = 1 \geq 1$.
- $B = (1, 0)$ satisfies $1^2 + 0 = 1 \geq 1$.
- Midpoint $M = (0.5, 0.5)$ gives $0.5^2 + 0.5 = 0.25 + 0.5 = 0.75 < 1$, so $M \notin$ the upper contour set.

Hence the upper contour set is not convex, so u is not quasi-concave on $\mathbb{R}^2_+$.

Caution:

- Sum of concave functions is concave.
- Sum of quasi-concave functions need not be quasi-concave.
- Always verify quasi-concavity directly or via recognized sufficient conditions.

3.3 Lagrange Method

Assume u is continuously differentiable, preferences are strictly monotonic, and u is quasi-concave. Consider the utility maximization problem:

$$\max_{x \in \mathbb{R}^n_+} u(x) \quad \text{s.t.} \quad p \cdot x \leq I.$$

Use a Lagrangian with multiplier $\lambda \geq 0$:

$$\mathcal{L} = u(x) - \lambda(p \cdot x - I).$$

For interior candidates (all $x_i > 0$), the FOCs are:

$$\frac{\partial \mathcal{L}}{\partial x_i} = \frac{\partial u(x^*)}{\partial x_i} - \lambda^* p_i = 0 \quad (i = 1, \ldots, n),$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = p \cdot x^* - I = 0.$$

Theorem 3.3.1 (Sufficiency of the Lagrangian Method)**.** *If u is continuously differentiable, strictly monotonic, and quasi-concave, and (x^*, λ^*) solves the first-order system with $(p, I) \in \mathbb{R}^n_{++} \times \mathbb{R}_+$, then x^* solves the consumer problem.*

Remark. Preferences are strictly (strongly) monotonic iff their utility representations are strictly (strongly) increasing.

Remark. Under strict monotonicity, the budget binds. Under quasi-concavity, the FOCs are sufficient (not just necessary).

From the FOCs at an interior optimum x^*,

$$\frac{\partial u(x^*)/\partial x_i}{p_i} = \frac{\partial u(x^*)/\partial x_j}{p_j} = \lambda^* \quad \forall i, j. \qquad \text{(Bang-for-Buck Equalization)}$$

Interpretation: the marginal utility per dollar is the same across all goods. Equivalent two-good statement:

$$MRS_{12}(x^*) = \frac{p_1}{p_2}.$$

- Interior optimum: tangency between the indifference curve (IC) and the budget line at x^*.
- Strict monotonicity pushes the optimum to the boundary, i.e. the budget is exhausted (the constraint binds).

Theorem 3.3.2. *If u is strictly quasi-concave, there is at most one solution.*

- Existence comes from continuity + compact budget set (Weierstrass).
- Uniqueness comes from strict quasi-concavity + convex feasible set.

Example 3.3.3. $u(x, y) = x + \sqrt{y}$, prices $(p_x, p_y) = (2, 4)$, income $I = 8$.

Set up the Lagrangian:

$$\mathcal{L}(x, y, \lambda) = x + \sqrt{y} + \lambda(8 - 2x - 4y).$$

FOCs:

$$\frac{\partial \mathcal{L}}{\partial x} = 1 - 2\lambda = 0, \qquad \frac{\partial \mathcal{L}}{\partial y} = \frac{1}{2\sqrt{y}} - 4\lambda = 0, \qquad \frac{\partial \mathcal{L}}{\partial \lambda} = 8 - 2x - 4y = 0.$$

Solve the FOCs (interior candidate):

- From $1 - 2\lambda = 0$: $\lambda^* = \frac{1}{2}$.
- From $\frac{1}{2\sqrt{y}} - 4\lambda = 0$: $\frac{1}{2\sqrt{y^*}} = 4 \cdot \frac{1}{2} = 2 \Rightarrow \sqrt{y^*} = \frac{1}{4} \Rightarrow y^* = \frac{1}{16}$.
- From $8 - 2x - 4y = 0$: $2x + 4y = 8 \Rightarrow x + 2y = 4 \Rightarrow x^* = 4 - 2 \cdot \frac{1}{16} = 4 - \frac{1}{8} = \frac{31}{8}$.

Check feasibility and assumptions (as the slide instructs):

- $x^* = \frac{31}{8} > 0$, $y^* = \frac{1}{16} > 0 \Rightarrow$ the candidate is interior and satisfies $x, y \geq 0$.
- Budget binds (also shown by the third FOC): $2x^* + 4y^* = 8$.
- u is continuously differentiable on $\{y > 0\}$ and the optimum has $y^* > 0$.
- u is strictly increasing in both goods on $\mathbb{R}^2_+$, and u is concave (hence quasi-concave), so the FOC method applies here.

The Lagrangian technique can fail (or require modifications) under:

- Non-differentiability (kinks).
- Corner solutions (some $x_i^* = 0$).
- Non-convex feasible sets.

- Non-continuous preferences/utility.

Practical advice: be slow, methodical, and geometric: draw ICs and the feasible set, then compare candidate bundles directly.

Perfect complements and perfect substitutes are key preference classes that often generate corner solutions, making interior FOCs uninformative.

- Perfect complements: kinked indifference curves (kinked ICs). Not differentiable.
- Perfect substitutes: linear indifference curves (linear ICs). Not strictly quasi-concave.

Both can place optima at corners, where interior FOCs are not informative.

Example 3.3.4. $u(x, y) = \min\{x, y\} + x + y$. This utility combines different structural components:

- It combines complementarity and substitutability components.
- Its ICs are piecewise linear with kinks and changing slopes.

Recall the Lexicographic Preferences on $\mathbb{R}^2_+$:

$$x \succeq y \text{ if either } x_1 > y_1, \text{ or } (x_1 = y_1 \text{ and } x_2 \geq y_2).$$

Key properties:

- Complete and transitive, but not continuous.
- No continuous utility representation on $\mathbb{R}^2_+$.

Thus, standard smooth optimization tools do not apply here.

Note that "more x_1" dominates any change in x_2. Optimal choice under a standard budget: spend to maximize x_1 first (the optimum lies at the feasible point with the largest x_1 on the budget).

3.4 Summary

- Existence: continuity + non-empty compact budget set.
- The budget set is compact and convex under standard assumptions ($p \in \mathbb{R}^n_{++}$, $I \geq 0$).
- Convex preferences $\iff$ quasi-concave utility representation.
- Strict quasi-concavity gives uniqueness.
- Lagrangian FOCs deliver bang-for-buck equalization and budget exhaustion in smooth interior cases.
- When assumptions fail (kinks, corners, non-convexities, lexicographic), use direct geometric/methodical analysis.

Checklist for Solving U-Max Problems

1. Write the feasible set clearly (including non-negativity).
2. Check continuity and compactness for existence.

3. Check monotonicity and (strict) quasi-concavity.
4. Use FOCs only when smooth interior assumptions are justified.
5. If not, inspect corners/kinks and compare utility directly.

4 Demand and Consumer Welfare

4.1 Marshallian Demand

Recall from last section. Utility maximization problem: Given prices $\boldsymbol{p}$ and income I, the consumer solves

$$\max\{u(\boldsymbol{x}) : \boldsymbol{x} \in B(\boldsymbol{p}, I)\}, \qquad B(\boldsymbol{p}, I) = \{\boldsymbol{x} \in \mathbb{R}^n_+ : \boldsymbol{p} \cdot \boldsymbol{x} \leq I\}.$$

- Under continuity and compactness, a solution exists.
- Under strict quasi-concavity, the solution is unique.

Because the solution to the utility-maximization problem may or may not be unique, we define

Definition 4.1.1. Define the demand correspondence as the set of optimal bundles:

$$X(\boldsymbol{p}, I) = \{\boldsymbol{x} \in B(\boldsymbol{p}, I) : \boldsymbol{x} \succeq \boldsymbol{y} \text{ for all } \boldsymbol{y} \in B(\boldsymbol{p}, I)\}.$$

This is called Marshallian demand correspondence.

It maps $(\boldsymbol{p}, I) \in \mathbb{R}^n_{++} \times \mathbb{R}_+$ into a set of optimal bundles. This set can have one point, many points, or (without assumptions) can fail to be well-defined.

To move from a correspondence to a function:

- If $X(\boldsymbol{p}, I)$ is a singleton, write $\boldsymbol{x}(\boldsymbol{p}, I)$ (the Marshallian demand function).
- If $X(\boldsymbol{p}, I)$ is not a singleton, keep the set notation $X(\boldsymbol{p}, I)$.

This distinction is important in examples like perfect substitutes with tied prices, where multiple optimal bundles can occur.

Example 4.1.2 (Perfect Substitutes)**.** Let $u(x_1, x_2) = x_1 + x_2$. Then the demand correspondence is

$$X(\boldsymbol{p}, I) = \begin{cases} \{(I/p_1, 0)\}, & p_1 < p_2, \\ \{(0, I/p_2)\}, & p_2 < p_1, \\ \{\boldsymbol{x} \in \mathbb{R}^2_+ : \boldsymbol{p} \cdot \boldsymbol{x} = I\}, & p_1 = p_2. \end{cases}$$

The tie case ($p_1 = p_2$) produces many optimizers on the budget line.

Building on the idea that some preferences generate unique demands while others may not, the following canonical examples summarize when demand is unique:

Example 4.1.3 (Canonical Demand Examples)**.** 1. Cobb–Douglas:

$$u(\boldsymbol{x}) = \prod_{j=1}^{n} x_j^{\alpha_j}, \qquad \alpha_j > 0.$$

Demands are unique:

$$x_i(\boldsymbol{p}, I) = \frac{\alpha_i I}{k p_i} \qquad \text{where } k = \sum_i \alpha_i.$$

2. Perfect complements:

$$u(\boldsymbol{x}) = \min\{x_1, \ldots, x_n\}.$$

Demands are unique:

$$x_i(\boldsymbol{p}, I) = \frac{I}{\sum_j p_j}.$$

3. Perfect substitutes:

$$u(\boldsymbol{x}) = \sum_j x_j.$$

Demands are not unique if there is more than one lowest price.

Definition 4.1.4 (Quasi-Linear Utility)**.** A useful class that often restores uniqueness and simplifies income effects is quasi-linear utility:

$$u(\boldsymbol{x}) = v(x_1, \ldots, x_{n-1}) + x_n.$$

Technically, perfect substitutes fit this definition.

Standard regularity conditions: v is twice differentiable, increasing, concave (often strictly). Then preferences are convex; often strictly convex with strict concavity of v. In this case, demands are unique and demand for x_1 is independent of income if income is sufficiently high.

Definition 4.1.5. For $f : X \subseteq \mathbb{R}^n \to \mathbb{R}$:

- Increasing:

$$\boldsymbol{x} \geq \boldsymbol{y} \Rightarrow f(\boldsymbol{x}) \geq f(\boldsymbol{y}).$$

- Strictly increasing:

$$\boldsymbol{x} > \boldsymbol{y} \Rightarrow f(\boldsymbol{x}) > f(\boldsymbol{y}).$$

- Strongly increasing:

$$\boldsymbol{x} \geq \boldsymbol{y} \text{ and } x_j > y_j \text{ for some } j \Rightarrow f(\boldsymbol{x}) > f(\boldsymbol{y}).$$

Remark. Strongly increasing function $\subset$ Strictly increasing function $\subset$ Increasing function.

The distinction between strict and strong only matters in multi-dimensional domains.

Definition 4.1.6. For $\succeq$ on $\mathbb{R}^n_+$,

- Strict monotonicity: If $x_i \geq y_i$ for all i then $\boldsymbol{x} \succeq \boldsymbol{y}$, and also if $x_i > y_i$ for all i then $\boldsymbol{x} \succ \boldsymbol{y}$.

(More of everything is strictly better.)

- Strong monotonicity: If $x_i \geq y_i$ for all i and $x_j > y_j$ for some j, then $\boldsymbol{x} \succ \boldsymbol{y}$. (More is strictly better.)

Remark. Axiom 4 from Lecture 2 is strong monotonicity. With complete preferences, it implies strict monotonicity.

Theorem 4.1.7. *If preferences $\succeq$ over $X \subseteq \mathbb{R}^n$ are represented by a utility function u, then $\succeq$ are strictly (strongly) monotonic if and only if u is strictly (strongly) increasing over X.*

Definition 4.1.8 (Homogeneity of degree k)**.** A function $f : X \to \mathbb{R}$ with $X \subseteq \mathbb{R}^n$ is homogeneous of degree k iff

$$f(t\boldsymbol{x}) = t^k f(\boldsymbol{x}) \quad \text{for all } t > 0 \text{ and all } \boldsymbol{x} \in X.$$

- Degree 1: doubling all arguments doubles the value.
- Degree 0: scale-invariant.
- Example: perfect substitutes/complements are degree 1.

Example 4.1.9 (Cobb–Douglas Homogeneity)**.** Consider Cobb–Douglas utility:

$$u(\boldsymbol{x}) = \prod_{j=1}^{n} x_j^{\alpha_j}, \qquad \alpha_j > 0 \text{ for all } j = 1, \ldots, n.$$

Let

$$\alpha_1 + \cdots + \alpha_n = k \qquad \text{for some } k > 0.$$

For $t > 0$,

$$u(t\boldsymbol{x}) = \prod_{j=1}^{n} (tx_j)^{\alpha_j} = \left(\prod_{j=1}^{n} t^{\alpha_j}\right)\left(\prod_{j=1}^{n} x_j^{\alpha_j}\right) = t^k u(\boldsymbol{x}).$$

Hence, Cobb–Douglas utility is homogeneous of degree

$$k = \alpha_1 + \alpha_2 + \cdots + \alpha_n.$$

Macro people like to use this as a production function where

$$\alpha_1 + \alpha_2 + \cdots + \alpha_n = 1.$$

Example 4.1.10 (Quasi-linear (non-)homogeneity)**.**

$$u(\boldsymbol{x}) = \sqrt{x_1} + x_2$$

is not homogeneous of any degree (most functions are not). How to see it is not: fix $t = 4$.

First, $u(1, 1) = 2$ and

$$u(4\mathbf{(1,1)}) = u(4, 4) = \sqrt{4} + 4 = 6 = 3u(1, 1).$$

If it were homogeneous of degree k, then $4^k = 3$.

- However, $u(4,1) = 3$ and

$$u(4(\mathbf{4,1})) = u(16,4) = \sqrt{16} + 4 = 8 = \frac{8}{3}u(4,1).$$

This suggests that if it were homogeneous of degree k, then $4^k = \frac{8}{3}$.

Since $3 \neq \frac{8}{3}$, this utility function cannot be homogeneous of any degree k.

Lemma 4.1.11. *Assume preferences on $\mathbb{R}^n_+$ are represented by a utility function u that is continuous and strongly monotonic. Then for all $(\boldsymbol{p}, I) \in \mathbb{R}^n_{++} \times \mathbb{R}_+$, the demand correspondence $X(\boldsymbol{p}, I)$ satisfies:*

1. *Non-empty.*
2. *Budget exhaustion:*
$$\boldsymbol{p} \cdot \boldsymbol{x} = I \qquad \text{for all } \boldsymbol{x} \in X(\boldsymbol{p}, I).$$
3. *Homogeneous of degree 0 in $(\boldsymbol{p}, I)$:*
$$X(\boldsymbol{p}, I) = X(t\boldsymbol{p}, tI), \qquad t > 0.$$
4. *Continuity under uniqueness: If demand is unique for each $(\boldsymbol{p}, I)$, then the Marshallian demand function $\boldsymbol{x}(\boldsymbol{p}, I)$ is continuous.*
   - *Note: when it is not a singleton set, the demand correspondence is upper semi-continuous / hemi-continuous (not needed for this class).*
5. *Convexity under quasi-concavity: If u is quasi-concave, then $X(\boldsymbol{p}, I)$ is a convex set.*
6. *Uniqueness under strict quasi-concavity: If u is strictly quasi-concave, then $X(\boldsymbol{p}, I)$ is a singleton (uniqueness).*

Why degree-0 homogeneity in $(\boldsymbol{p}, I)$:

Scaling all prices and income by $t > 0$ leaves the budget set unchanged. Therefore the optimizing bundles are unchanged: there is no money illusion in the model (the budget line for $(\boldsymbol{p}, I)$ is the same as for $(t\boldsymbol{p}, tI)$).

4.2 Indirect Utility

Definition 4.2.1 (Indirect Utility Function)**.** The indirect utility function is defined by

$$v(\boldsymbol{p}, I) = \max\{u(\boldsymbol{x}) : \boldsymbol{x} \in \mathbb{R}^n_+,\ \boldsymbol{p} \cdot \boldsymbol{x} \leq I\}.$$

If demand is unique, then

$$v(\boldsymbol{p}, I) = u(\boldsymbol{x}(\boldsymbol{p}, I)).$$

It summarizes welfare level as a function of the market environment $(\boldsymbol{p}, I)$.

Example 4.2.2 (Cobb–Douglas Indirect Utility)**.** For Cobb–Douglas utility

$$u(\boldsymbol{x}) = \prod_j x_j^{\alpha_j}, \qquad \alpha_j > 0,$$

optimal demand is

$$x_j(\boldsymbol{p}, I) = \frac{\alpha_j}{k}\frac{I}{p_j} \qquad \text{with } k = \sum_k \alpha_k.$$

Plugging back yields

$$v(\boldsymbol{p}, I) = \prod_j \left(\frac{\alpha_j}{k}\frac{I}{p_j}\right)^{\alpha_j} = \left(\frac{I}{k}\right)^k \prod_j \left(\frac{\alpha_j}{p_j}\right)^{\alpha_j}.$$

Notice: Goods are normal. Income up $\Rightarrow$ demand, utility up. Opposite for prices.

Demands are quite general and allowed to go up, down, or stay the same with prices and income. Recall: substitutes, complements, normal and inferior goods. For indirect utility we will get some general properties regarding prices and income. For that, we need two more definitions.

Definition 4.2.3 (Quasi-Convexity)**.** For $f : \mathbb{R}^n_+ \to \mathbb{R}$, f is quasi-convex iff

$$f(\boldsymbol{x}) \geq f(\boldsymbol{y}) \;\Rightarrow\; f(t\boldsymbol{x} + (1-t)\boldsymbol{y}) \leq f(\boldsymbol{x}), \qquad t \in [0,1].$$

In words, averages are worse than extremes. In quasi-convexity, the average must be worse than the higher extreme, while in quasi-concavity, the average must be better than the lower extreme.

Theorem 4.2.4. *A function f is quasi-convex if and only if $-f$ is quasi-concave.*

A quasi-convex function has convex inferior (worse-than) sets.

Theorem 4.2.5. *If f is convex, then f is quasi-convex.*

Just as quasi-concavity is related to concavity of a function, quasi-convexity is related to convexity of a function, not to convexity of a set.

Example 4.2.6. Linear indifference curves indicate that perfect substitute utility functions are both quasi-convex and quasi-concave.

Example 4.2.7. For $u(x, y) = x^2 + y$, indifference curves $u(x, y) = c$ are

$$y = c - x^2,$$

downward-opening parabolas. The lower contour set $\{(x, y) : x^2 + y \leq c\} = \{(x, y) : y \leq c - x^2\}$ is the region below a concave curve, which is convex, so u is quasi-convex. The upper contour set $\{(x, y) : x^2 + y \geq c\} = \{y \geq c - x^2\}$ is generally not convex, so u is not quasi-concave.

Definition 4.2.8. For $f : X \subseteq \mathbb{R}^n \to \mathbb{R}$, decreasing is defined as

$$\boldsymbol{x} \geq \boldsymbol{y} \Rightarrow f(\boldsymbol{x}) \leq f(\boldsymbol{y}).$$

Remark. Analogous definitions for strictly and strongly decreasing functions.

Theorem 4.2.9 (Properties of Indirect Utility)**.** *If u is continuous and strongly monotonic, then the indirect utility $v(\boldsymbol{p}, I)$ is:*

1. *continuous on $\mathbb{R}^n_{++} \times \mathbb{R}_+$,*
2. *homogeneous of degree 0,*
3. *strictly increasing in I,*
4. *decreasing in $\boldsymbol{p}$,*
5. *quasi-convex in $(\boldsymbol{p}, I)$.*

Theorem 4.2.10 (Roy's Identity)**.** *If v is differentiable at $(\boldsymbol{p}, I)$ and $\partial v/\partial I \neq 0$, then*

$$x_i(\boldsymbol{p}, I) = -\frac{\partial v(\boldsymbol{p}, I)/\partial p_i}{\partial v(\boldsymbol{p}, I)/\partial I}, \qquad i = 1, \ldots, n.$$

This is convenient to recover Marshallian demand from indirect utility by differentiation.

4.3 Dual Problem: Expenditure Minimization

Definition 4.3.1 (Expenditure Function)**.** Given prices $\boldsymbol{p}$ and a utility target $\bar{u}$, define the expenditure function as

$$e(\boldsymbol{p}, \bar{u}) = \min_{\boldsymbol{x} \in \mathbb{R}^n_+} \{\boldsymbol{p} \cdot \boldsymbol{x} : u(\boldsymbol{x}) \geq \bar{u}\}.$$

The solution $\boldsymbol{x}^h(\boldsymbol{p}, \bar{u})$ is the Hicksian (compensated) demand.

It gives the minimal cost to reach the utility target $\bar{u}$ at prices $\boldsymbol{p}$.

To find $\boldsymbol{x}^h(\boldsymbol{p}, \bar{u})$ and $e(\boldsymbol{p}, \bar{u})$:

- Fix the utility contour $\bar{u}$.
- Slide the expenditure line inward until the first tangency.
- The tangency point gives $\boldsymbol{x}^h(\boldsymbol{p}, \bar{u})$ and $e(\boldsymbol{p}, \bar{u})$, with

$$e(\boldsymbol{p}, \bar{u}) = \boldsymbol{p} \cdot \boldsymbol{x}^h(\boldsymbol{p}, \bar{u}).$$

Finding the expenditure function: Solving the e-min problem can be done in the differentiable case using Lagrangian-type methods. Under standard assumptions of quasi-concave utility, strong monotonicity, and an interior solution:

- it satisfies Bang-for-Buck equalization (MRS equals the price ratio); and
- utility is just equal to $\bar{u}$ (the constraint binds).

Theorem 4.3.2 (Properties of the Expenditure Function)**.** *If u is continuous and strictly increasing, then the expenditure function $e(\boldsymbol{p}, \bar{u})$ is:*

1. 0 *when* $\bar{u} = u(0)$,
2. *continuous on its domain,*
3. *strictly increasing and unbounded above in the utility level* $\bar{u}$,
4. *increasing in* $\boldsymbol{p}$,
5. *homogeneous of degree* 1 *in* $\boldsymbol{p}$,
6. *concave in* $\boldsymbol{p}$.

Theorem 4.3.3 (Duality Relations)**.** *If u is continuous and strictly increasing, then*

$$e(\boldsymbol{p}, v(\boldsymbol{p}, I)) = I, \qquad v(\boldsymbol{p}, e(\boldsymbol{p}, \bar{u})) = \bar{u}.$$

Theorem 4.3.4 (Marshallian–Hicksian Link)**.** *If u is strictly increasing, strictly quasiconcave, and continuous, then*

$$\boldsymbol{x}(\boldsymbol{p}, I) = \boldsymbol{x}^h(\boldsymbol{p}, v(\boldsymbol{p}, I)), \qquad \boldsymbol{x}^h(\boldsymbol{p}, \bar{u}) = \boldsymbol{x}(\boldsymbol{p}, e(\boldsymbol{p}, \bar{u})).$$

4.4 Consumer Welfare

Recall from first year micro.

$$\text{Consumersurplus} = \text{area under demand} - \text{total expenditure}.$$

Welfare change is often proxied by ΔCS.

Remark. This is exact only under special conditions (notably quasi-linear utility / negligible income effects).

General welfare analysis asks a money-metric question: How much money are you willing to pay to move from situation A to situation B?

This can be triggered by:

- Price changes,
- Income changes,
- Any policy/environment change affecting utility.

Example 4.4.1. Take utility

$$u(x_1, x_2) = \ln x_1 + x_2,$$

with budget

$$p_1 x_1 + p_2 x_2 \leq I, \qquad x_1, x_2 \geq 0.$$

We study welfare effects of a change in p_1.

Step 1 (FOCs and corner logic)

Set up the Lagrangian:

$$\mathcal{L} = \ln x_1 + x_2 + \lambda\,(I - p_1 x_1 - p_2 x_2).$$

For an interior solution with $x_2 > 0$, the FOCs are

$$\frac{1}{x_1} = \lambda p_1, \qquad 1 = \lambda p_2 \Rightarrow x_1 = \frac{p_2}{p_1}.$$

Then

$$x_2 = \frac{I - p_1 x_1}{p_2} = \frac{I}{p_2} - 1,$$

which requires $I \geq p_2$ for feasibility ($x_2 \geq 0$).

Step 2 (Piecewise form): full Marshallian demand

Combining the interior-case solution with the corner case, the full Marshallian demand is:

$$x_1(p_1, p_2, I) = \begin{cases} \frac{I}{p_1}, & I < p_2, \\ \frac{p_2}{p_1}, & I \geq p_2, \end{cases} \qquad x_2(p_1, p_2, I) = \begin{cases} 0, & I < p_2, \\ \frac{I}{p_2} - 1, & I \geq p_2. \end{cases}$$

For $I \geq p_2$, demand for good 1 is income-independent. This is the no-income-effect feature used in the welfare result.

Fix I and p_2. In the high-income region $I \geq p_2$,

$$x_1(p_1) = \frac{p_2}{p_1} \qquad (\text{equivalently } p_1 = \frac{p_2}{x_1}).$$

Fix $I = 80$ and $p_2 = 40$. The price of good 1 changes from $p_1 = 40$ to $p_1' = 20$.

Since $I \geq p_2$, we are in the high-income region where

$$x_1(p_1) = \frac{p_2}{p_1}.$$

Thus

$$x_1 = \frac{40}{40} = 1, \qquad x_1' = \frac{40}{20} = 2.$$

$$\Delta CS = \int_{p_1'}^{p_1} x_1(p_1)\,dp_1 = \int_{20}^{40} \frac{40}{p_1}\,dp_1 = 40 \ln 2 \approx 27.73.$$

Rectangle + triangle approximation gives about 30 (slight overestimate).

Initial situation $(p_1, p_2, I) = (40, 40, 80)$:

$$(x_1, x_2) = (1, 1), \qquad u = \ln 1 + 1 = 1.$$

After the price drop to $p_1' = 20$:

$$(x_1', x_2') = (2, 1).$$

Now ask: how much income m can be taken away at new prices while keeping utility at 1?

At new prices $(20, 40)$, with income reduced to $80 - m$:

$$x_1' = 2 \qquad \text{(unchanged; no income effect for good 1).}$$

$$x_2' = \frac{80 - m}{40} - 1 = 1 - \frac{m}{40}.$$

So

$$u' = \ln 2 + 1 - \frac{m}{40}.$$

Equalize to old utility $u = 1$:

$$\ln 2 + 1 - \frac{m}{40} = 1 \Rightarrow m = 40 \ln 2.$$

Hence willingness to pay equals $40 \ln 2$, exactly matching ΔCS.

In this quasi-linear setup, good 1 has no income effect in the relevant region. So the area under demand captures the exact money-metric welfare change. Result in this example:

$$\Delta CS = CV = 40 \ln 2.$$

Definition 4.4.2 (Compensating Variation (CV))**.** For a change from $(\boldsymbol{p}, I)$ to $(\boldsymbol{p}', I')$, with

$$u = u(\boldsymbol{x}(\boldsymbol{p}, I)), \qquad u' = u(\boldsymbol{x}(\boldsymbol{p}', I')),$$

define

$$CV = e(\boldsymbol{p}', u') - e(\boldsymbol{p}', u) = I' - e(\boldsymbol{p}', u).$$

It measures how much income can be taken away at new prices while leaving the person at old utility.

To interpret compensating variation (CV):

- $CV > 0$: the new situation is better (the consumer is willing to pay for it).
- $CV < 0$: the new situation is worse (the consumer needs compensation).

A sign-convention note is important here: JR defines CV with the opposite sign. These slides use the more common convention in which positive CV means there has been a welfare gain.

Remark. CV is the correct money-metric welfare measure. By contrast, ΔCS is exact only when income effects are negligible, as in the quasi-linear benchmark. When income effects are substantial, ΔCS can either overstate or understate the true welfare change.

Why ΔCS can be biased: The reason is that ΔCS is obtained by integrating Marshallian demand. Marshallian demand contains both substitution effects and income effects. Therefore, if marginal willingness to pay changes strongly with wealth, the area-under-demand measure becomes biased as a welfare measure.

Definition 4.4.3 (Equivalent Variation)**.** An alternative money-metric welfare measure is Equivalent Variation (EV), defined as

$$EV = e(\boldsymbol{p}, u') - e(\boldsymbol{p}, u).$$

This uses old prices $\boldsymbol{p}$ and new utility u'.

This contrasts with Compensating Variation (CV):

- EV uses old prices $\boldsymbol{p}$ and new utility u'.
- CV uses new prices $\boldsymbol{p}'$ and old utility u.

Both EV and CV are theoretically valid money-metric welfare measures.

Putting the main ideas together:

- Demand is naturally a correspondence; uniqueness requires curvature, specifically strict quasi-concavity.
- Demand has a robust structure: it is non-empty, satisfies budget exhaustion, and is homogeneous of degree 0 in $(\boldsymbol{p}, I)$.
- Indirect utility is continuous, increasing in income, decreasing in prices, and quasi-convex.
- The utility-maximization problem (U-max) and the expenditure-minimization problem (E-min) are dual to each other; this duality relies on continuity and monotonicity assumptions.
- In welfare analysis, CV and EV are money-metric measures, while consumer surplus (CS) is exact in the quasi-linear benchmark and only approximate otherwise.

5 Uncertainty and Expected Utility

Uncertainty appears in many familiar situations.

Example 5.0.1. The contents of a box of chocolate are uncertain, the realization of exam questions is uncertain, an opponent's behavior in a game is uncertain, the financial payoff from risky assets is uncertain, and the outcomes of lottery tickets are uncertain.

These examples show that many economic decisions are made under uncertainty rather than under certainty.

In previous weeks, the alternatives were bundles $x \in \mathbb{R}^n_+$. We studied preferences over bundles, denoted by $\succeq$, and then obtained a utility representation. The consumer's problem was to maximize utility over the budget set. However, we now move from certain outcomes to uncertain outcomes. Because of this shift, we need a new set of alternatives, we need axioms that are appropriate for that set, and we need a representation that is consistent with those axioms.

This leads to three guiding questions:

1. What is the right set of alternatives under uncertainty?
2. What axioms should preferences satisfy on that set?
3. What representation form should utility take?

Under uncertainty, the primitive alternatives are not certain bundles anymore. Instead, let

$$Z = \{z_1, \ldots, z_n\}$$

be the set of outcomes. In this course, Z is assumed to be finite, although the results can be extended to continuous probability distributions. The elements of Z may be bundles of goods, amounts of money, or complete descriptions of a situation.

Therefore, the relevant set of alternatives is the set of lotteries L over outcomes in Z, and individual lotteries are denoted by lowercase letters such as p and q.

Definition 5.0.2. A lottery is a probability distribution over Z. Formally,

$$p = (p_1, \ldots, p_n), \qquad p_i \geq 0, \qquad \sum_{i=1}^{n} p_i = 1.$$

Accordingly, the set of all lotteries is

$$L = \Delta(Z) = \{p \in \mathbb{R}^n_+ : \sum_i p_i = 1\}.$$

This means that preferences under uncertainty are defined over lotteries, that is, $\succeq$ is defined on L.

When we want to describe a lottery over only part of the outcome set, we can write it in the form

$$(z_1, p_1; \cdots ; z_n, p_n).$$

Example 5.0.3. $(z_1, 0.2; z_2, 0.8)$ means that z_1 occurs with probability 20% and z_2 occurs with probability 80%. This notation makes the probabilistic structure of the alternative explicit.

The geometry of the lottery set is also important. When $n = 3$, the set L is a triangle, or equivalently a 2-simplex. The lecture emphasizes that this set is compact, convex, and a subset of $\mathbb{R}^3_+$. In the diagram, the three vertices are labeled $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$, and the point p represents one particular lottery in the set.

At this point, the lecture briefly contrasts two different frameworks for decision under uncertainty.

1. In the expected utility framework studied in this lecture, the primitive objects are lotteries over outcomes, so a lottery is written as
$$p : Z \to [0, 1].$$

2. By contrast, in subjective expected utility, which is outside the scope of the course here, the primitive objects are acts that map states to outcomes, written as
$$f : S \to Z.$$

This contrast matters because the two theories use different primitive objects and therefore require different axiomatizations. At the same time, they share a similar representation logic. In the subjective expected utility framework, preferences over acts also reveal the decision maker's subjective beliefs. Thus, the lecture is clarifying that today's analysis is about lotteries over outcomes, not acts over states.

5.1 St. Petersburg and Expected Value

The St. Petersburg setup asks the decision maker to choose between two alternatives. Option A is to receive \$1000 for sure. Option B is a gamble in which a coin is tossed until the first tail appears on toss n, and the payoff is 2^n. What criterion should describe choice?

Definition 5.1.1. For a lottery p, expected value is defined as

$$EV(p) = \sum_{z \in Z} p_z z.$$

The associated rule is simple: choose the lottery with the higher expected value. This gives a direct way to rank lotteries when outcomes can be evaluated numerically.

Example 5.1.2. Suppose lottery p gives a gain of 100 with probability 1/2 and a loss of 50 with probability 1/2. Then

$$EV(p) = 0.5(100) + 0.5(-50) = 25.$$

Now compare this with lottery q, which gives 40 for sure. In that case,

$$EV(q) = 40.$$

Since $40 > 25$, the expected-value criterion ranks q above p.

In the St. Petersburg game, to receive payoff 2^n, the coin must land heads on the first $n-1$ tosses and tails on the n-th toss. Because the flips are independent, the probability of this event is

$$2^{-n}.$$

Therefore, the expected value of option B is

$$EV(B) = \sum_{n=1}^{\infty} 2^{-n} \cdot 2^n = \sum_{n=1}^{\infty} 1 = \infty.$$

So, under the expected-value criterion, option B has infinite expected value.

According to expected value, one should choose option B over any finite sure amount. However, behaviorally, most people prefer some finite sure amount instead. This mismatch between the prediction of expected value and observed choice motivates replacing expected value with expected utility.

Bernoulli's resolution begins by identifying what the expected-value criterion assumes about utility. If choices are based only on expected monetary value, then utility must effectively be linear in wealth, that is,

$$u(w) = w.$$

Bernoulli argued instead that the marginal value of wealth diminishes. He expressed this by

$$\frac{du}{dw} = \frac{1}{w} \quad \Rightarrow \quad u(w) = \ln w.$$

Therefore, lotteries should be evaluated by $E[\ln w]$ rather than by $E[w]$.

Applying this idea to the St. Petersburg game changes the ranking of the two options. For option A, which gives \$1000 for sure,

$$EU(A) = \ln(1000).$$

For option B, the payoff is 2^n when the first tail appears on toss n, which occurs with probability 2^{-n}. Hence

$$EU(B) = \sum_{n=1}^{\infty} 2^{-n} \ln(2^n) = \sum_{n=1}^{\infty} \frac{n \ln 2}{2^n} = 2 \ln 2 = \ln 4.$$

This calculation uses the fact that

$$\sum_{n=1}^{\infty} \frac{n}{2^n} = 2.$$

Since

$$\ln(1000) > \ln 4 \quad \Rightarrow \quad EU(A) > EU(B).$$

So under logarithmic utility, the sure \$1000 is preferred to the St. Petersburg gamble.

The paradox disappears once utility is concave. Concavity captures diminishing marginal utility of wealth, and this makes expected utility a more plausible criterion than expected value for describing actual behavior under risk.

5.2 Expected Utility Criterion

Now we generalize the idea into the full expected utility framework.

Definition 5.2.1. Let

$$Z = \{z_1, \ldots, z_n\}$$

be the set of outcomes. A person is an expected utility maximizer if there exists a utility function

$$u : Z \to \mathbb{R}$$

such that

$$p \succeq q \iff \sum_{i=1}^{n} p_i u(z_i) \geq \sum_{i=1}^{n} q_i u(z_i).$$

We say that u represents the preference relation $\succeq$ over the lottery set L.

This means that lotteries are ranked by the expected value of utility across outcomes, rather than by expected monetary value alone.

Rewrites this representation more formally by defining

$$V(p) = \sum_{i=1}^{n} p_i u(z_i).$$

With this notation,

$$p \succeq q \iff V(p) \geq V(q).$$

This makes clear that the object directly representing preferences over lotteries is V, since V is defined on lotteries, while u is defined only on certain outcomes. Still, under expected utility, V is linear in probabilities, so knowing u is equivalent to knowing V. This is why the lecture can sometimes speak loosely of u representing preferences over lotteries.

Example 5.2.2 (Simple Numerical Example)**.** The outcomes are

$$z_1 = (1, 1), \qquad z_2 = (1, 2), \qquad z_3 = (3, 0),$$

so that

$$Z = \{z_1, z_2, z_3\}.$$

Take the utility function

$$u(x_1, x_2) = x_1 x_2.$$

Then the utility of each outcome is

$$u(z_1) = 1, \qquad u(z_2) = 2, \qquad u(z_3) = 0.$$

Now consider the lottery

$$p = \left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right).$$

Its expected utility is

$$EU(p) = \frac{1}{2} \cdot 1 + \frac{1}{3} \cdot 2 + \frac{1}{6} \cdot 0 = \frac{7}{6}.$$

So the example shows exactly how expected utility is computed: first evaluate each outcome with u, then take the probability-weighted average.

Receiving z_1 for sure corresponds to the lottery $(1, 0, \ldots, 0)$. Thus, certainty is embedded in the same object class as risk, rather than being treated separately. More generally, any certain outcome z_i can be represented by the degenerate lottery that assigns probability 1 to z_i and probability 0 to all other outcomes.

Definition 5.2.3. A compound lottery is a lottery over lotteries.

This means that uncertainty can arise in multiple stages, since one first faces uncertainty over which lottery is selected and then uncertainty within the selected lottery itself. The expected utility framework handles this by reducing compound lotteries to simple lotteries.

Example 5.2.4. With probability 1/2, one draws lottery p, and with probability 1/2, one draws lottery q.

The reduction principle says that a compound lottery such as

$$\frac{1}{2}p + \frac{1}{2}q$$

can be represented as a simple lottery whose final outcome probabilities are weighted averages of the probabilities in p and q. The lecture's example is

$$\frac{1}{2}\left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right) + \frac{1}{2}(1, 0, 0) = \left(\frac{3}{4}, \frac{1}{6}, \frac{1}{12}\right).$$

Coordinate by coordinate, this works as follows:

$$\frac{1}{2}\left(\frac{1}{2},\frac{1}{3},\frac{1}{6}\right)=\left(\frac{1}{4},\frac{1}{6},\frac{1}{12}\right), \qquad \frac{1}{2}(1,0,0)=\left(\frac{1}{2},0,0\right),$$

and adding them gives

$$\left(\frac{1}{4},\frac{1}{6},\frac{1}{12}\right)+\left(\frac{1}{2},0,0\right)=\left(\frac{3}{4},\frac{1}{6},\frac{1}{12}\right).$$

So the reduced simple lottery assigns probability 3/4 to the first outcome, 1/6 to the second, and 1/12 to the third.

5.3 Axioms for EU Representation

The central question is: why can we not simply reuse all the axioms from the certainty model? The answer is that the environment has changed. The alternatives are now lotteries, not bundles, so some axioms transfer naturally while others do not.

L, the set of lotteries, is still a large subset of $\mathbb{R}^n_+$,

- so continuity remains a natural requirement.
- However, strong monotonicity has little bite in the simplex, because probabilities must sum to 1, so increasing one probability necessarily decreases another.
- In addition, convexity restrictions can be too strong in this context. The reason is that the alternatives are lotteries rather than outcomes, and a convex combination of lotteries is simply a compound lottery.

Therefore, the practical route is not to reuse all certainty axioms mechanically, but instead to start from a choice criterion—expected utility—and derive the axioms that identify it.

Because $L \subset \mathbb{R}^n_+$ is closed and convex, if preferences on L are complete, transitive, and continuous, then they admit a continuous utility representation on L.

However, expected utility requires more than mere continuity. Specifically, it requires linearity in probabilities. So a general continuous utility representation is not yet enough to obtain the expected utility form.

Axiom 5.3.1 (A1: Completeness and Transitivity)**.** Completeness means that any two lotteries are comparable. Transitivity means that rankings are internally consistent.

These are baseline rationality axioms, playing the same role here as they do in the certainty model.

Axiom 5.3.2 (A2: Continuity)**.** For each lottery $p \in L$, the upper and lower contour sets are closed:

$$G(p)=\{q \in L : q \succeq p\}, \qquad W(p)=\{q \in L : p \succeq q\}.$$

This axiom rules out abrupt jumps in preferences. An equivalent intuition: if z_1 is the worst certain outcome and z_n is the best certain outcome, then for any lottery l, there exists some probability $p \in [0,1]$ such that

$$l \sim (z_1, p; z_n, 1-p).$$

So every lottery can be matched, in terms of preference, by an appropriate mixture of the worst and best certain outcomes.

Axiom 5.3.3 (A3: Independence)**.** For all $p, q, r \in L$ and all $\alpha \in (0, 1)$,

$$p \succeq q \iff \alpha p + (1-\alpha)r \succeq \alpha q + (1-\alpha)r.$$

This means that mixing both alternatives with the same third lottery preserves their ranking. If p is preferred to q, then adding the same background risk r in the same proportion should not reverse that preference. The lecture notes that JR splits this axiom into two parts, substitution and reduction.

Theorem 5.3.4 (Expected Utility Theorem)**.** *preferences over lotteries satisfy A1, A2, and A3 if and only if they admit an expected utility representation. Formally,*

$$p \succeq q \iff \sum_i p_i u(z_i) \geq \sum_i q_i u(z_i).$$

If two utility functions u and v represent the same expected utility preferences, then they must satisfy

$$u = \alpha v + \beta, \qquad \alpha > 0.$$

So an expected utility representation is unique only up to a positive affine transformation.

This is stronger than the certainty case, because ordinary ordinal utility over bundles is preserved under any strictly increasing transformation, whereas expected utility allows only affine changes.

The implication is that expected utility imposes much more structure than a generic ordinal representation.

5.4 Risk Attitudes

Let $Z \subseteq \mathbb{R}$, so outcomes are money levels.

Definition 5.4.1. A person is

1. strictly risk averse if they prefer the sure amount equal to the lottery's expected value to any risky lottery with the same mean;
2. risk neutral if they are indifferent between the risky lottery and the sure amount with the same mean;
3. strictly risk loving if they prefer the risky lottery to the sure amount with the same mean.

For an expected utility maximizer,

- strictly concave u is equivalent to strict risk aversion
- linear u is equivalent to risk neutrality
- and strictly convex u is equivalent to strict risk loving.

Thus, risk attitude is fully encoded in the curvature of utility.

For a concave utility function, the utility of the mean wealth is at least as large as the mean of utility, and for a strictly concave utility function with a non-degenerate lottery, the inequality is strict:

$$u(E[w]) \geq E[u(w)],$$

with

$$u(E[w]) > E[u(w)]$$

under strict concavity and non-degeneracy. This is the Jensen intuition. It explains why a risk-averse person prefers the sure amount equal to the lottery's expected value rather than the risky lottery itself.

For a lottery p, the certainty equivalent CE is defined by

$$u(CE) = EU(p),$$

so CE is the sure amount that makes the person indifferent to the lottery. The risk premium is

$$\pi = EV(p) - CE.$$

Under strict risk aversion,

$$\pi > 0.$$

Equivalently, a strictly risk-averse person has

$$CE < EV(p),$$

meaning they would accept less than the expected value in exchange for certainty.

Example 5.4.2. Consider a lottery that yields $w_H = 100$ and $w_L = 0$, each with probability 1/2. Its expected value is

$$EV = \frac{1}{2} \cdot 100 + \frac{1}{2} \cdot 0 = 50.$$

Now suppose utility is given by

$$u(w) = \sqrt{w}.$$

Then expected utility is

$$EU = \frac{1}{2}\sqrt{100} + \frac{1}{2}\sqrt{0} = 5.$$

The certainty equivalent CE is the sure amount that gives the same utility as the lottery, so it is defined by

$$u(CE) = EU.$$

Since $u(w) = \sqrt{w}$, this means

$$\sqrt{CE} = 5,$$

and therefore

$$CE = 25.$$

The risk premium is then

$$\pi = EV - CE = 50 - 25 = 25.$$

This example shows concretely that with a concave utility function, the certainty equivalent lies below expected value, so the risk premium is positive.

5.5 Summary

Under uncertainty, the primitive alternatives are lotteries over outcomes. The expected-value criterion is not sufficient, because it fails in the St. Petersburg case. The expected utility criterion instead ranks lotteries by the expected utility of outcomes. The axioms A1, A2, and A3 characterize when such an expected utility representation exists. That representation is unique only up to positive affine transformations. Finally, when outcomes are money levels, the curvature of utility maps directly into risk attitude.

Putting these pieces together, the lecture's main message is that choice under uncertainty requires a different framework from choice under certainty. The correct objects are lotteries, not bundles. Expected value is too crude because it ignores diminishing marginal utility. Expected utility provides the right representation, its axioms explain exactly when it applies, and its curvature properties explain why different people may be risk averse, risk neutral, or risk loving. The two-outcome example at the end confirms all of this numerically in one place.

6 Producer Theory

This week we will discuss producer theory. We focus on costs and production. Next lecture we focus on profit maximization with perfectly competitive firms.

Cost minimization is one important part of profit maximization. Indeed, to maximize profit, you must be minimizing the cost of what you produce.

There are many similarities between consumer and producer theory. There are also differences. Production numbers have cardinal meaning, while utility numbers do not. Also, there is not typically any income constraint for production.

A firm acquires inputs and combine them to produce output.

- Inputs are purchased on input markets and these expenditures are the firm's costs.
- Outputs are sold on product markets and the firm earns revenue form these sales.

Assumption: firms maximise profit.

The firm has a production possibility set, $Y \subset R^m$, where each vector $\mathbf{y} = (y_1, \cdots, y_m) \in Y$ is a production plan; $y_i < 0$ if resource i is an input; $y_i > 0$ if resource i is an output.

If a firm produces a single output y from a vector of inputs $\mathbf{x} = (x_1, \cdots, x_n)$, with $\mathbf{x} \geq 0$, and $y \geq 0$, the production function, f, is a mapping from $\mathbb{R}^n_+$ into $\mathbb{R}_+$.

$$y = f(\mathbf{x})$$

means that y units of output (and no more) can be produced using the input vector $\mathbf{x}$.

Definition 6.0.1 (Assumption: Properties of the Production Function)**.** The production function, $f : \mathbb{R}^n_+ \to \mathbb{R}_+$, is continuous, strictly increasing, and strictly quasiconcave on $\mathbb{R}^n_+$, and $f(0) = 0$.

Example 6.0.2. 2 inputs (can easily extend to $n \geq 1$ inputs)

- 1. Perfect Complements: $f(x_1, x_2) = \min\{x_1, x_2\}$

- 2. Perfect Substitutes: $f(x_1, x_2) = x_1 + x_2$
- 3. Cobb-Douglas: $f(x_1, x_2) = x_1^\alpha x_2^\beta$

Definition 6.0.3. A production function $f(x)$ has:

(1) Constant returns to scale iff $f(tx) = tf(x)$ for all $t > 0$;

(2) Decreasing returns to scale iff $f(tx) < tf(x)$ for all $t > 1$;

(3) Increasing returns to scale iff $f(tx) > tf(x)$ for all $t > 1$;

Let $y = f(x)$ be a production function, where $x \in \mathbb{R}^n_+$. Define the production set $Y = \{(-x, y) : y \leq f(x) \text{ and } x \in \mathbb{R}^n_+\}$. Production sets are useful for general equilibrium analysis:

- inputs for some firms are outputs of others
- inputs are negative, outputs are positive

One condition imposed on production sets for existence of a general equilibrium is convexity of the production set.

Theorem 6.0.4. *The production set $Y = \{(-x, y) : y \leq f(x) \text{ and } x \in \mathbb{R}^n_+\}$ is a convex set if and only if $f(x)$ is a concave function.*

We assume that firms are perfectly competitive on their input markets and face fixed input prices. $\mathbf{w} = (w_1, \cdots, w_n) \geq 0$ is the vector of market prices of inputs $\mathbf{x} = (x_1, \cdots, x_n)$. The firm maximises profit; hence it will choose the least costly vector of inputs capable of producing y.

Definition 6.0.5 (The Cost Function)**.** The cost function, defined for all input prices $\mathbf{w} \gg 0$ and all input levels $y \in f(\mathbb{R}^n_+)$ is the minimum-value function,

$$c(\mathbf{w}, y) \equiv \min_{\mathbf{x} \in \mathbb{R}^n_+} \mathbf{w} \cdot \mathbf{x} \quad \text{s.t.} \quad f(\mathbf{x}) \geq y.$$

If $\mathbf{x}(\mathbf{w}, y)$ solves the cost-minimisation problem, then

$$c(\mathbf{w}, y) = \mathbf{w} \cdot \mathbf{x}(\mathbf{w}, y).$$

Because f is strictly increasing, the constraint will always be binding at a solution. Consequently, the cost-minimisation problem is equivalent to

$$\min_{\mathbf{x} \in \mathbb{R}^n_+} \mathbf{w} \cdot \mathbf{x} \quad \text{s.t.} \quad y = f(\mathbf{x}) \tag{3.1}$$

Because $\mathbf{w} \gg 0$ and f is strictly quasiconcave, the solution $\mathbf{x}^* \equiv \mathbf{x}(\mathbf{w}, y)$ to (3.1) is unique; it is called the firm's conditional input demand, because it is conditional on the level of output y, which need not be profit maximising.

For simplicity, assume $\mathbf{x}^* \gg 0$ and that f is differentiable at $\mathbf{x}^*$ with $\nabla f(\mathbf{x}^*) \gg 0$. By Lagrange's theorem, there is a $\lambda^* \in \mathbb{R}$ such that

$$w_i = \lambda^* \frac{\partial f(\mathbf{x}^*)}{\partial x_i}, \qquad i = 1, \cdots, n.$$

Since $w_i > 0$, $i = 1, \cdots, n$, we may divide the preceding ith equation by the jth to obtain

$$\frac{\partial f(\mathbf{x}^*)/\partial x_i}{\partial f(\mathbf{x}^*)/\partial x_j} = \frac{w_i}{w_j} \tag{3.2}$$

Thus, cost minimisation implies that the marginal rate of substitution between any two inputs is equal to the ratio of their prices.

Example 6.0.6 (2 inputs)**.** 1. Perfect Complements: $f(x_1, x_2) = \min\{x_1, x_2\}$. Cost Function: $c(\mathbf{w}, y) = (w_1 + w_2)y$

2. Perfect Substitutes: $f(x_1, x_2) = x_1 + x_2$. Cost Function: $c(\mathbf{w}, y) = \min\{w_1, w_2\}y$.

3. Cobb-Douglas: $f(x_1, x_2) = x_1^\alpha x_2^\beta$. Cost Function:

$$c(\mathbf{w}, y) = \left[\left(\frac{\alpha}{\beta}\right)^{\frac{\beta}{\alpha+\beta}} + \left(\frac{\alpha}{\beta}\right)^{\frac{-\alpha}{\alpha+\beta}}\right] w_1^{\frac{\alpha}{\alpha+\beta}} w_2^{\frac{\beta}{\alpha+\beta}} y^{\frac{1}{\alpha+\beta}}.$$

- If $\alpha + \beta = 1$, then cost is linear (constant marginal cost): $c(\mathbf{w}, y) = Ay$ for some constant A.
- If $\alpha + \beta = \frac{1}{2}$, then cost is quadratic (linear increasing marginal cost): $c(\mathbf{w}, y) = Ay^2$ for some constant A.

Theorem 6.0.7 (Properties of the Cost Function)**.** *If f is continuous and strictly increasing, then $c(\mathbf{w}, y)$ is*

1. *Zero when $y = 0$,*
2. *Continuous on its domain,*
3. *For all $\mathbf{w} \gg 0$, is strictly increasing and unbounded above in y,*
4. *Increasing in $\mathbf{w}$,*
5. *Homogeneous of degree one in $\mathbf{w}$,*
6. *Concave in $\mathbf{w}$,*
7. *Shephard's lemma: $c(\mathbf{w}, y)$ is differentiable in $\mathbf{w}$ at $(\mathbf{w}^0, y^0)$ whenever $\mathbf{w}^0 \gg 0$, and*

$$\frac{\partial c(\mathbf{w}^0, y^0)}{\partial w_i} = x_i(\mathbf{w}^0, y^0), \qquad i = 1, \cdots, n.$$

The firm's cost-minimisation problem is similar to the consumer's expenditure-minimisation problem; duality between production and cost just as between utility and expenditure:

- If we begin with a production function and derive its cost function, we can take that cost function and use it to generate a production function
- If the original production function is quasiconcave, the derived production function will be identical to it. If the original production function is not quasiconcave, the derived production function is a 'concavication' of it.

Any function with all the properties of a cost function generates some production function for which it is the cost function.

Theorem 6.0.8 (Recovering a Production Function from a Cost Function)**.** *Let* $c : \mathbb{R}^n_{++} \times \mathbb{R}_+ \to \mathbb{R}_+$ *satisfy properties 1 to 7 of a cost function given in Theorem 6.0.7. Then the function* $f : \mathbb{R}^n_+ \to \mathbb{R}_+$ *defined by*

$$f(\mathbf{x}) \equiv \max\{y \geq 0 \mid \mathbf{w} \cdot \mathbf{x} \geq c(\mathbf{w}, y),\ \forall \mathbf{w} \gg 0\}$$

is increasing, unbounded above, and quasiconcave. Moreover, the cost function generated by f *is* c*.*

The short run is typically defined as a period where some of the inputs are fixed.The long run is typically defined as the period where all inputs can be changed and possibly firms can enter and exit.

From the cost curves we can obtain average and marginal cost curves in the the standard manner:

$$AC(y) = \frac{c(\mathbf{w}, y)}{y} \quad \text{and} \quad MC(y) = \frac{\partial c(\mathbf{w}, y)}{\partial y}.$$

Summary

- Producer theory mimics in some ways consumer theory.
- The production has cardinal meaning unlike utilities.
- The cost function has the same types of properties as the expenditure function.
- There are various ways to represent production and technology.
- We focused mainly on a production function.
- Other ways are to just give the cost function, or a production set which is a more general concept for more complicated technologies that are not separable.

7 Competitive Firms

Last lecture, we talked about cost and production. We focused on the cost minimization problem. This lecture, we focus on profit maximization with perfectly competitive firms.

Cost minimization is one important part of profit maximization. Indeed, to maximize profit, you must be minimizing the cost of what you produce. Today we will the profit maximization problem for perfectly competitive firms.

When the firm is both a perfect competitor on input and on its output market, it is a price taker on both output and input markets.

We suppose the firm's objective is to maximise profits. A firm that produces a single output chooses the level of output and the combination of inputs that solve:

$$\max_{(\mathbf{x},y)\geq 0} \ py - \mathbf{w} \cdot \mathbf{x} \quad \text{s.t.} \quad f(\mathbf{x}) \geq y. \tag{3.6}$$

where $f(\mathbf{x})$ is a production function satisfying Assumption 3.1.

We may replace the inequality in the constraint by an equality because the production function is strictly increasing and rewrite the profit maximisation problem as:

$$\max_{\mathbf{x}\in\mathbb{R}^n_+} pf(\mathbf{x}) - \mathbf{w}\cdot\mathbf{x}. \tag{3.7}$$

Assume this problem has an interior solution $\mathbf{x}^* \gg 0$. Then the profit-maximising amount of output is $y^* = f(\mathbf{x}^*)$ and the first-order conditions require that the gradient of the maximand be zero because there are no constraints; that is,

$$p\frac{\partial f(\mathbf{x}^*)}{\partial x_i} = w_i, \quad \text{for every } i = 1, \ldots, n.$$

When f satisfies Assumption 6.0.1 and is in addition strictly concave, solutions to the maximisation problem (3.6) – when they exist – are unique for each price vector $(p, \mathbf{w})$.

The optimal choice of output, $y^* \equiv y(p, \mathbf{w})$ is called the firm's output supply function.

The optimal choice of input, $\mathbf{x}^* \equiv \mathbf{x}(p, \mathbf{w})$, gives the firm's input demand functions.

Unlike the conditional input demands that depend on output, these input demands only depend on prices; they maximise the firm's profit.

Definition 7.0.1 (The Profit Function)**.** The firm's profit functions depends only on input and output prices and is defined as the maximum-value function,

$$\pi(p, \mathbf{w}) \equiv \max_{(\mathbf{x},y)\geq 0} py - \mathbf{w}\cdot\mathbf{x} \quad \text{s.t.} \quad f(\mathbf{x}) \geq y.$$

Example 7.0.2. Cobb Douglass: $y = f(\mathbf{x}) = x_1^\alpha x_2^\beta$ with $\alpha, \beta > 0$ and $\alpha + \beta < 1$.

The assumption that $\alpha + \beta < 1$ is to make the production function strictly concave (on $\mathbb{R}^2_{++}$). How do we know? (See tutorial 7 question) We can show that general Cobb Douglass is quasi-concave on $\mathbb{R}^2_+$ and strictly quasi-concave on $\mathbb{R}^2_{++}$. These results extend to $n \geq 1$ goods.

Here after substituting $f(\mathbf{x})$ for y into $\pi(y, \mathbf{x}) = py - (w_1x_1 + w_2x_2)$, we get:

$$\pi(\mathbf{x}) = px_1^\alpha x_2^\beta - (w_1x_1 + w_2x_2).$$

As the function $f(\mathbf{x})$ is differentiable and strictly concave, we can look for a solution using the first order conditions (FOC's):

$$\frac{\partial \pi(\mathbf{x})}{\partial x_i} = 0 \quad \text{for } i = 1, 2.$$

This yields:

$$p\alpha x_1^{\alpha-1}x_2^\beta = w_1 \tag{1}$$

$$p\beta x_1^\alpha x_2^{\beta-1} = w_2 \tag{2}$$

These two (FOC's) imply Bang 4 Buck Equalization:

$$\frac{\dfrac{\partial \pi(\mathbf{x})}{\partial x_1}}{w_1} = \frac{\dfrac{\partial \pi(\mathbf{x})}{\partial x_2}}{w_2}.$$

Solving the profit max for Cobb Douglas:

Recall the F0C's:

$$p\alpha x_1^{\alpha-1}x_2^{\beta} = w_1 \quad (1) \qquad p\beta x_1^{\alpha}x_2^{\beta-1} = w_2 \quad (2)$$

Also, the production function:

$$y = x_1^{\alpha}x_2^{\beta} \tag{3}$$

From (3) we observe that:

$$x_1^{\alpha-1}x_2^{\beta} = \frac{y}{x_1} \quad \text{and} \quad x_1^{\alpha}x_2^{\beta-1} = \frac{y}{x_2}.$$

Substituting (3) into (1) and (2) are re-arranging, we find:

$$x_1 = \frac{\alpha p}{w_1}y \quad \text{and} \quad x_2 = \frac{\beta p}{w_2}y. \tag{4}$$

These are not finished as we want $x_i(p, \mathbf{w})$.

Solving the profit max for Cobb Douglas (cont.):

$$y = x_1^{\alpha}x_2^{\beta} \tag{3}$$

$$x_1 = \frac{\alpha p}{w_1}y \quad \text{and} \quad x_2 = \frac{\beta p}{w_2}y \tag{4}$$

Substituting (4) back into (3) we can solve for y, then substitute it back into (4) to get x_1 and x_2. Round and round we go...

Step 1, sub (4) into (3):

$$y = x_1^{\alpha}x_2^{\beta} = \left(\frac{\alpha p}{w_1}y\right)^{\alpha}\left(\frac{\beta p}{w_2}y\right)^{\beta}.$$

Step 2, collect y terms:

$$y^{1-(\alpha+\beta)} = \left(\frac{\alpha p}{w_1}\right)^{\alpha}\left(\frac{\beta p}{w_2}\right)^{\beta}.$$

Step 3, power each side to $\dfrac{1}{1-(\alpha+\beta)}$ and rearrange to get:

$$y(p, \mathbf{w}) = \left[\left(\frac{\alpha}{w_1}\right)^{\alpha}\left(\frac{\beta}{w_2}\right)^{\beta}p^{\alpha+\beta}\right]^{\left(\frac{1}{1-(\alpha+\beta)}\right)}. \tag{5}$$

Solving the profit max for Cobb Douglas (cont. (cont.)):

$$y(p, \mathbf{w}) = \left[\left(\frac{\alpha}{w_1}\right)^{\alpha}\left(\frac{\beta}{w_2}\right)^{\beta}p^{\alpha+\beta}\right]^{\left(\frac{1}{1-(\alpha+\beta)}\right)} \tag{5}$$

$$x_1 = \frac{\alpha p}{w_1}y \quad \text{and} \quad x_2 = \frac{\beta p}{w_2}y \tag{4}$$

Step 4, Substitute (5) into x_1 of (4) to get:

$$x_1 = \left(\frac{\alpha p}{w_1}\right)\left[\left(\frac{\alpha}{w_1}\right)^{\alpha}\left(\frac{\beta}{w_2}\right)^{\beta}p^{\alpha+\beta}\right]^{\left(\frac{1}{1-(\alpha+\beta)}\right)}$$

and rearrange to get:

$$x_1(p, \mathbf{w}) = \left[\left(\frac{\alpha}{w_1}\right)^{1-\beta} \left(\frac{\beta}{w_2}\right)^{\beta} p \right]^{\left(\frac{1}{1-(\alpha+\beta)}\right)}. \tag{6}$$

Step 5, Substitute (5) in to x_2 of (4) to get:

$$x_2(p, \mathbf{w}) = \left[\left(\frac{\alpha}{w_1}\right)^{\alpha} \left(\frac{\beta}{w_2}\right)^{1-\alpha} p \right]^{\left(\frac{1}{1-(\alpha+\beta)}\right)}. \tag{7}$$

Note:

$$y(p, \mathbf{w}) = \left[\left(\frac{\alpha}{w_1}\right)^{\alpha} \left(\frac{\beta}{w_2}\right)^{\beta} p^{\alpha+\beta} \right]^{\left(\frac{1}{1-(\alpha+\beta)}\right)} \quad \text{and} \quad x_1 = \frac{\alpha}{w_1} py \quad \text{and} \quad x_2 = \frac{\beta}{w_2} py \quad (4)$$

So:

$$\begin{aligned} \pi(p, \mathbf{w}) &= py - (w_1 x_1 + w_2 x_2) \\ &= py - \left(w_1 \left(\frac{\alpha}{w_1}\right) py + w_2 \left(\frac{\beta}{w_2}\right) py \right). \end{aligned}$$

So:

$$\pi(p, \mathbf{w}) = (1 - (\alpha + \beta))py.$$

So:

$$\pi(p, \mathbf{w}) = (1 - (\alpha + \beta)) \left[p \left(\frac{\alpha}{w_1}\right)^{\alpha} \left(\frac{\beta}{w_2}\right)^{\beta} \right]^{\left(\frac{1}{1-(\alpha+\beta)}\right)}.$$

Clearly profit is increasing in output price p and decreasing in input prices w. What else?

Example 7.0.3. Fix $w_1 = 1$, $w_2 = 2$, $p = 10$, $\alpha = \frac{1}{2}$, $\beta = \frac{1}{4}$.

Then, profit is:

$$\pi(10, (1, 2)) = \left(\frac{1}{4}\right) \left[10 \left(\frac{1}{2}\right)^{\frac{1}{2}} \left(\frac{1}{8}\right)^{\frac{1}{4}} \right]^4 = \frac{10000}{128} = 78.125.$$

What happens to profit if we double all prices?

$$\pi(20, (2, 4)) = \left(\frac{1}{4}\right) \left[(2)10 \left(\frac{1}{(2)2}\right)^{\frac{1}{2}} \left(\frac{1}{(2)8}\right)^{\frac{1}{4}} \right]^4 = 2\pi(10, (1, 2)).$$

So it seems to be homogeneous of degree 1 in prices.

Theorem 7.0.4. *If the production function is continuous, strictly increasing and strictly concave and $f(0) = 0$, then for all non-negative $(p, \mathbf{w})$, the profit function $\pi(p, \mathbf{w})$ is continuous and*

- *1. Increasing in p.*
- *2. Decreasing in $\mathbf{w}$.*
- *3. Homogeneous of degree 1 in $(p, \mathbf{w})$.*

- *4. Convex in* $(p, \mathbf{w})$.

Theorem 7.0.5 (Properties of Output Supply and Input Demand Functions)**.** *Suppose that f is a strictly concave production function satisfying Assumption 3.1 and that its associated profit function, $\pi(p, y)$, is twice continuously differentiable. Then, for all $p > 0$ and $\mathbf{w} \gg 0$ where it is well defined:*

- *1. Homogeneity of degree zero:*

$$y(tp, t\mathbf{w}) = y(p, \mathbf{w}) \quad \textit{for all } t > 0$$

$$x_i(tp, t\mathbf{w}) = x_i(p, \mathbf{w}) \quad \textit{for all } t > 0 \textit{ and } i = 1, \ldots, n.$$

- *2. Own-price effects:*

$$\frac{\partial y(p, \mathbf{w})}{\partial p} \geq 0,$$

$$\frac{\partial x_i(p, \mathbf{w})}{\partial w_i} \leq 0 \quad \textit{for all } i = 1, \ldots, n.$$

Back to our example of Cobb Douglas.

Special cases: $\alpha + \beta = \dfrac{1}{2}$. Then notice that the level of y is:

$$y(p, \mathbf{w}) = \left[\left(\frac{\alpha}{w_1} \right)^{\alpha} \left(\frac{\beta}{w_2} \right)^{\beta} p^{\alpha+\beta} \right]^{\left(\frac{1}{1-(\alpha+\beta)} \right)} = kp \quad \text{where } k > 0 \text{ is a constant based on } \alpha, \beta, \mathbf{w}.$$

In other words supply is linear through the origin. Recall that the supply curve is the Marginal cost curve, so this is the case of linear marginal cost. In the previous lecture we showed this case corresponds to the cost curve $c(\mathbf{w}, y) = Ay^2$ where $A > 0$ is also a constant based on $\alpha, \beta, \mathbf{w}$. These are all connected.

What if $\alpha + \beta \geq 1$?

The solutions from the profit max were based on $\alpha + \beta < 1$ so we cannot use them here. However, we can still use the cost function from Lecture 6.

$$c(\mathbf{w}, y) = \left[\left(\frac{\alpha}{\beta} \right)^{\frac{\beta}{\alpha+\beta}} + \left(\frac{\alpha}{\beta} \right)^{\frac{-\alpha}{\alpha+\beta}} \right] w_1^{\frac{\alpha}{\alpha+\beta}} w_2^{\frac{\beta}{\alpha+\beta}} y^{\frac{1}{\alpha+\beta}}.$$

The derivation of the cost function for the Cobb-Douglass case only uses the fact that the production function is strictly quasi-concave.

Case: $\alpha + \beta = 1$: Then

$$c(\mathbf{w}, y) = \left[\left(\frac{\alpha}{\beta} \right)^{\beta} + \left(\frac{\alpha}{\beta} \right)^{-\alpha} \right] w_1^{\alpha} w_2^{\beta} y = dy$$

where $d > 0$ is a constant.

Hence: profit $\pi = (p - d)y$ where $p, d > 0$ are constants.

3 cases:

(a) (Price is too high) ($p > d$): Profit is increasing in y and unbounded above. So NO SOLUTION.

(b) (Price is too low) ($p < d$): Profit is negative for all $y > 0$, and zero for $y = 0$, the unique solution is $y = 0$ and $\mathbf{x} = (0, 0)$.

(c) (Price is just right) ($p = d$): Every pair of supply y and corresponding cost minimizing inputs $\mathbf{x}(\mathbf{w}, y)$ form a solution for all $y \geq 0$.

(Wonderful field of macro dreams and constant returns to scale, if you build it....they will come...)

Case of $\alpha + \beta > 1$.

In this case, the cost function becomes of the form:

$$c(\mathbf{w}, y) = dy^{\delta} \quad \text{where } \delta = \frac{1}{\alpha + \beta} < 1 \text{ is a constant and } d > 0 \text{ is a constant.}$$

So,

$$\frac{\partial^2 c(\mathbf{w}, y)}{\partial y^2} = (\delta - 1)\delta d y^{\delta - 1} < 0 \quad \text{as } 0 < \delta < 1.$$

In other words, MC is declining in y.

So, there will be NO SOLUTION.....to the profit max problem.
Profit will be unbounded above....

Example 7.0.6. If $p = \alpha = \beta = w_1 = w_2 = 1$, then $c(\mathbf{w}, y) = \sqrt{y}$. So profit $\pi = py - c(\mathbf{w}, y) = y - \sqrt{y}$ which is strictly convex. It starts at zero, and profit falls below zero until $y = 1/4$, then in increases without bound....Hence, there is no profit maximizing solution

MAIN TAKE AWAY: Perfect competition and decreasing MC are typically not compatible.

Short Run: When some inputs are fixed.

Example 7.0.7. $f(\mathbf{x}) = x_1^{\frac{1}{2}} x_2^{\frac{1}{4}}$ but fix $x_2 = 16$, $w_2 = 2$.

So, short run production function is:

$$f(x_1, 2) = x_1^{\frac{1}{2}} (16)^{\frac{1}{4}} = 2\sqrt{x_1}.$$

$$\pi = p\, 2\sqrt{x_1} - (w_1 x_1 + 2(16)).$$

FOC:

$$\frac{\partial \pi}{\partial x_1} = 0 \;\Rightarrow\; x_1(p, \mathbf{w}) = \left(\frac{p}{w_1}\right)^2.$$

Now: This leads to

$$y(p, \mathbf{w}) = \frac{2p}{w_1}.$$

So:

$$\pi(p, \mathbf{w}) = \frac{p^2}{w_1} - 32.$$

In the example with $w_1 = 1$, $p = 10$, and the ability to adjust both inputs, we got:

$$\pi = 78.125.$$

Now, we get $\pi = 100 - 32 = 68$.

In general, profit must be lower in the short run. MAIN TAKE AWAY: Profit max problem simplifies, but optimal input mix is not obtained.

For producers, welfare is simply profit. We do not have to construct the willingness to pay. Conveniently everything is in dollars already.

Notice again differences between consumer theory and produce theory. In the end, welfare is what economists are interested in. "How much are you willing to pay (give up)?" is the right question. In the case of producers, the answer is simple due to the profit max assumption (LOSS in profit). For consumers it is more complicated: "The amount of income (wealth) needed to obtain the previous welfare level"

Summary

- The Profit Max can be done directly on the production function, or:
- It can be done by first minimizing cost for each output and then choosing output y to maximize profit.
- Next week we will put things together in competitive markets.
- Then shake it up with imperfect competition.
- Then go back to perfect competition, but with general equilibrium (explicitly interactive markets).

Part II
Markets and Welfares

8 Perfect Competition

This week we will talk about perfect competition. We will look at short and long run type results. Next week we look at imperfect competition more. Then we go to general equilibrium theory which is back to perfect competition with many markets.

Perfectly Competitive Markets: Buyers and sellers take prices as given. Market Demand is the sum of individual demands. Market Supply is the sum of individual supplies.

What is a perfectly competitive equilibrium? A perfectly competitive equilibrium in a market is a price p and and a quantity Y such that Market Supply equals Market Demand. Behind this is profit maximization by each seller and utility maximization by each buyer.

Example 8.0.1 (Perfectly Competitive Market)**.**

- Demand Side: 64 consumers each with the same utility function $u(y, z) = \ln y + z$. We are interested in the market for good y so we fix the price of good z at $p_z = 1$. We also presume the income of each consumer $I \geq 1$. Why?
- Supply Side: 4 Sellers each with the same production function

$$y = f(x_1, x_2) = x_1^{1/4} x_2^{1/4}.$$

We will also fix the input prices to be: $w_1 = 4$; $w_2 = 1$. We denote the price of good y simply by p.

Demand Side: Find demand $y(p)$ of each consumer by maximizing utility and sum the demands over the market to get the market demand curve.

$$u(y, z) = \ln y + z$$

Notice that this is quasi-linear, so no income effects for good y provided income is sufficiently high. U-max implies: $y^d = \frac{1}{p}$ where $p > 0$ is the price of y and $p_z = 1$ (is fixed).

To be able to afford this, we need that the cost of y which is $py^d = 1$. Hence, we assume that $I \geq 1$.

As there are 64 consumers and all are identical, we simply get the demand to be:

$$Y^D = 64y^d = \frac{64}{p}.$$

Market Demand is: $Y^D = \frac{64}{p}$.

Note, if we had some new consumers of Cobb-Douglas type with $u(y, z) = y^\alpha z^\beta$ then, each would have demand $y = \frac{A}{p}$ where $A > 0$ is a constant depending on α, β and income. These could easily be added to the Market Demand.

Definition 8.0.2. • The short run is a situation where some inputs are fixed and entry and exit are not allowed.

• The long run is a situation where all inputs are variable and entry and exit are allowed.

We consider both today, starting with the short run. In the short run, we will fix the 2nd input to be $x_2 = 16$. In the long run, we will allow all inputs to be variable and the number of sellers to be variable (free entry and exit). We will compare the outcomes of the two runs: short and long.

8.1 Short Run

For short run, we assume $y = f(x_1, x_2) = x_1^{\frac{1}{4}} x_2^{\frac{1}{4}}$ where $x_2 = 16$ (the 2nd input is fixed), and input prices are $w_1 = 4$; $w_2 = 1$. So short run production function is: $y = 2x_1^{\frac{1}{4}}$.

Profit is Rev – Cost: we have 2 ways to proceed:

1. (2 steps) Find Cost function $C(y)$ and then max profit $\pi(y)$ in terms of y and x_1.

2. (1 step) Directly find max $\pi(x_1, y)$ as a function of x_1, y.

I will follow (1).

Cost function is $C(y) = w_1x_1 + w_2x_2$ as the input markets are competitive. Recall, competitive means you can buy as much as you like at a fixed price, here, $w_1 = 4$ and $w_2 = 1$ are the fixed prices. Noting that $x_2 = 16$ is fixed, we have a fixed cost of $w_2x_2 = 16$. Thus, $C(y) = w_1x_1(y) + 16$.

We need to find the optimal level of x_1 as a function of y. As $y = 2x_1^{\frac{1}{4}}$ is a strictly increasing function of x_1 we can invert it to get:

$$x_1(y) = \frac{y^4}{16}$$

which is the input demand for x_1 to produce y. So, $C(y) = \dfrac{y^4}{4} + 16$.

Profit function is

$$\pi(y) = py - C(y) = py - \left(\frac{y^4}{4} + 16\right).$$

Notice that the revenue is linear, so the profit function is concave as long as the cost function is convex. Recall that the sum of concave functions is concave. Linear is concave and $-f(y)$ is concave if and only if $f(y)$ is convex. So, we can use the first order approach and look for a point where MR = MC which is equivalent to $\dfrac{d\pi}{dy} = 0$. Any such point will be a maximum.

In our case, the cost function is strictly convex, so the profit function is strictly concave in y. As such, there will be a unique profit maximizer. If you did not derive the cost function and used the information to obtain profit directly on the inputs, you will find that:

$$\pi(x_1) = p(2x_1^{\frac{1}{4}}) - 4x_1$$

As profit is clearly a concave function of x_1 and you can use the first order approach to find the optimal inputs and outputs. Both methods will yield the same results.

SOC: $\frac{d^2\pi}{dy^2} = -3y^2 \leq 0$ as $y \geq 0$ (so concave). $\frac{d\pi}{dy} = 0$ implies $y^s = p^{\frac{1}{3}}$. (one seller's supply as a function of price of output)

For input demand x_1, we simply plug $y = p^{\frac{1}{3}}$ into $x_1(y) = \frac{y^4}{16} = \frac{p^{\frac{4}{3}}}{16}$. As we are interested in the supply in the MARKET, we sum the individual seller supplies to get: $Y^s = 4p^{\frac{1}{3}}$ (4 identical sellers).

A short run competitive equilibrium is a price p and a quantity Y such that market supply equals market demand.

Recall that demand is $Y^D = \frac{64}{p}$ and supply is $Y^s = 4p^{\frac{1}{3}}$. So, we are looking for a solution to $\frac{64}{p} = 4p^{\frac{1}{3}}$. This is solved when $p = 8$. Then plugging this into either the supply or the demand equation, we get: $Y^D = Y^S = Y = 8$. Note: each seller provides $y^S = 2$, and each consumer buys $y^d = \frac{1}{8}$. Profit of each seller is:

$$\pi(y) = py - \left(\frac{y^4}{4} + 16\right) = 8 \cdot 2 - \left(\frac{2^4}{4} + 16\right) = -4$$

Still better to produce as shutting down would leave profit of -16.

8.2 Long Run

Long Run Equilibrium (free entry and all inputs are variable): A long run competitive equilibrium is price p, a quantity Y and a number of firms n such that:

(a) Demand equals supply

(b) Profit of each seller is zero.

Back to the example, for the long run equilibrium in the previous example, we will presume there is a fixed cost of entry which is $F = 4$. What will happen in the long run if we do not assume a fixed cost of entry?

Recall that the production is: $y = f(x_1, x_2) = x_1^{\frac{1}{4}} x_2^{\frac{1}{4}}$. Input prices are $w_1 = 4$; $w_2 = 1$. But now both x_1 and x_2 are free to take on any non-negative value.

From Lecture 6 know that absent entry costs, we have the cost:

$$c(w, y) = \left[\left(\frac{\alpha}{\beta}\right)^{\frac{\beta}{\alpha+\beta}} + \left(\frac{\alpha}{\beta}\right)^{\frac{-\alpha}{\alpha+\beta}}\right] w_1^{\frac{\alpha}{\alpha+\beta}} w_2^{\frac{\beta}{\alpha+\beta}} y^{\frac{1}{\alpha+\beta}}$$

In this example, $\alpha = \beta = \frac{1}{4}$, which after substituting in for the prices $w_1 = 1$ and $w_2 = 4$ gives: $c(y) = 4y^2$. Adding the fixed cost of entry of 4 we get: $C(y) = 4 + 4y^2$. This is the cost of each seller operating in the long run.

Long Run: $\pi(y) = py - C(y) = py - (4y^2 + 4)$. Again, this is concave, so we can use the first order condition. Profit max at an interior point will require $\frac{d\pi}{dy} = 0$. This is equivalent to $MR = MC$.

Noting that $MR = p$ in a competitive market $\frac{d\pi}{dy} = 0$ requires:

$$p = MC.$$

Also, for a Long Run Competitive equilibrium we need $\pi(y) = 0$, which translates in a competitive market to

$$p = AC.$$

So, we can look for a Long Run Competitive equilibrium by FIRST checking outputs where:

$$MC = AC.$$

Total Cost: $C(y) = 4 + 4y^2$. Marginal Cost $C'(y) = 8y$. Average Cost $AC(y) = \frac{C(y)}{y} = \frac{4+4y^2}{y} = \frac{4}{y} + 4y$. So, $MC = AC$ when: $8y = \frac{4}{y} + 4y$. That is, $y = 1$ is the unique long run equilibrium output of a seller. As $p = MC$ for profit max, we know that $p = C'(1) = 8$. So we are almost there, the last step is finding the number of sellers in the LRCE.

To get the equilibrium number of sellers, we use the equilibrium demand to determine the equilibrium Y (total output) at the equilibrium price $p = 8$.

$$Y = \frac{64}{p} = \frac{64}{8} = 8$$

The total number of sellers $n = \frac{8}{y}$ where $y = 1$ is the LRCE quantity per seller. Thus, $n = \frac{8}{1} = 8$. The long run equilibrium is summarized as: $Y = 8$, $p = 8$, $n = 8$.

- Q1: What if the entry cost was not there in the previous example, e.g., $C(y) = 4y^2$?
  A: Then, $p = AC = MC$ requires $p = y = 0$. Then there is no long run equilibrium. Demand is not well defined when $p = 0$.
- Q2: What if the equilibrium number of sellers is not an integer?
  A: The typical thing to do is to take the largest integer n that yields positive profit and let price adjust to where supply equals demand given that number of sellers. Each seller makes a small positive profit, but entry of one more would drive profit below zero.
- Q2: What if there are different types of sellers?
  A: Those with inferior technologies (higher average costs) will be driven out. There may be a mix of seller types.
- Q3: Do there ever exist multiple Long run equilibria?
  A: Yes. In the textbook they give an example. Example 4.3 (page 169-170) constant returns to scale.

9 Imperfect Competition: Monopoly

For imperfect competition we will look at monopoly today. Next lecture we will consider imperfect competition with oligopoly. This requires some understanding of game theory. The main difference between perfect competition and imperfect competition is market power of the sellers. This means we don't look for a price where supply equals demand.

Monopoly typically involves some barrier to entry from other sellers:

- Big start up cost
- Government restrictions
- Patent on technology.

In this case, the seller need not take price as given. So, price p as well as the inputs x_1, x_2 and output level y are choice variables for the seller. The consumers are then assumed to take the chosen price p as given and just consume according to their demand curve. More generally, a seller with such market power might use more complicated pricing, e.g., price discrimination, etc.

Example 9.0.1. Cost: $C(y) = 4 + 4y^2$. Demand: $Y^D = \dfrac{64}{p}$. The seller notices that they can choose output y and then to maximize profit, they will choose the p to be on the demand curve.

In other words, they will choose the pair (p, y) so that $y = \dfrac{64}{p}$. Thus, the profit can be expressed in terms of y alone. For this, we need to invert the demand curve to get $p(y) = \dfrac{64}{y}$. Thus, $\pi(y) = py - C(y) = \left(\dfrac{64}{y}\right) y - C(y) = 64 - (4 + 4y^2)$.

The profit function: $\pi(y) = 64 - (4 + 4y^2)$ is concave, so we can look for an interior solution using the FOC $\dfrac{d\pi}{dy} = 0$ which is also where MR = MC. However, in this example MR = 0 as Revenue=64 is constant. So technically, there is no interior solution. The solution of $y = 0$ is also not a solution as it leads to an undefined price (essentially infinite price). So, there is no solution.

What the seller would like to do is set a very high price and limit output. However, by raising price a bit more, he can generate a higher profit and approach a profit of 64 in the limit. In reality, the demand curve will have some maximum price and this will be set, with the corresponding demand as the quantity.

These problems are not always there, but we should keep in mind that sometimes there may not be a solution. Most likely in the previous example, there is some maximum price that consumers will pay. The monopolist will then choose that price and realize the resulting profit.

Example 9.0.2 (A less problematic example)**.** Demand: $Y^D = 40 - \dfrac{p}{2}$ linear demand.

$C(y) = 4 + 4y^2$.

Invert demand to get $p(y) = 80 - 2y$. So profit is $\pi(y) = (80 - 2y)y - (4 + 4y^2)$. This is concave (second derivative $\dfrac{d^2\pi}{dx^2} = -12 < 0$ and has a profit maximizing $y = 6.66$.

Comparison of Monopoly and Perfect Competition

- Prices are typically higher in monopoly
- Quantities are typically lower in monopoly
- Welfare of producers is at least as high in monopoly
- Welfare of consumers is typically lower in monopoly

- Total welfare = consumer surplus + producer surplus is typically lower in monopoly
- Sometimes, it is more efficient to have a monopoly, e.g., decreasing Average Cost across relevant market demand levels.

10 Imperfect Competition: Oligopoly

This week we look at imperfect competition beyond monopoly. In particular we will look at Oligopoly. We will look at short run and long run oligopoly.

An oligopoly involves multiple sellers, more than 1 and not so many as to obtain perfect competition. Like monopoly, each seller has market power. There are various models of oligopoly. I will focus on quantity setting varieties.

10.1 Cournot

Example 10.1.1 (Duopoly)**.** Like a monopoly situation, we typically start with a demand curve which describes the market. We might have linear demand given as $D(p) = 100 - 2p$. We also need some description of production, technology or costs. The standard way to proceed is to start with the cost function. $C_i(q_i) = (q_i)^2$ where the subscript i denotes the seller 1 or 2. In this environment, is it natural to ask what will happen in equilibrium.

The standard approach to this situation is to look for an equilibrium treating the market as a game between the two sellers. The equilibrium concept for games is known as a Nash equilibrium which here will be a pair of quantities q_1, q_2 that maximize the profit of each seller given the quantity chosen by the other seller.

In general, a Nash equilibrium in an n-player game is an n-tuple of strategies $s = (s_1, \cdots, s_n)$ such that each player's strategy maximizes his utility given the strategies of the other players. In a quantity setting model of oligopoly, the sellers strategies are quantities.

In more general models, the strategies might include choices of quantities, prices, advertising, technology, lobbying governments, information acquisition, etc.

Based on the information given, we can form the profit function of each seller which can be taken to be the utility function of the seller.

First, invert the demand curve since the choices will be quantities. Recall that this can always be done due to the law of demand (monotonic). From $q(p) = 100 - 2p$, we find: $p(q) = 50 - \frac{q}{2}$.

Next, note that the total quantity q is the sum of the individual seller quantities, that is $q = q_1 + q_2$. Then, we form the profit functions:

$$\pi_1(q_1, q_2) = p(q)q_1 - C_1(q_1) \quad \text{and} \quad \pi_2(q_1, q_2) = p(q)q_2 - C_2(q_2)$$

We see that the profits are interrelated here by the quantity choices. Technically, this is a situation of an externality.

For seller 1:

$$\pi_1(q_1, q_2) = \left(50 - \frac{q_1 + q_2}{2}\right) q_1 - (q_1)^2$$

For seller 2:

$$\pi_2(q_1, q_2) = \left(50 - \frac{q_1 + q_2}{2}\right) q_2 - (q_2)^2$$

A Nash equilibrium is a pair (q_1, q_2) such that:

1. $\pi_1(q_1, q_2) \geq \pi_1(q_1', q_2)$ for all $q_1' \geq 0$
2. $\pi_2(q_1, q_2) \geq \pi_2(q_1, q_2')$ for all $q_2' \geq 0$

If there are more sellers, we just get more profit max conditions – one for each seller.

The solutions can be found using calculus in the typical cases which involve differentiable profit functions. Then we should check the profit functions are concave to use the first order conditions.

There are results about when this will be met, they include things like: concave demand functions (in one's own choice variables) and convex cost functions (in one's own choice variables). In those cases we can typically also get existence of an Nash equilibrium.

Sometimes existence of an equilibrium may fail. For the simple examples we will look at, existence holds. Existence results are often based on fixed point arguments. A Nash equilibrium can be viewed as a fixed point of a specific correspondence. Then we can apply a fixed point theorem from math for existence.

Check that each seller's profit function is concave it the seller's strategy. Then, find where the FOC holds for each seller. Use these FOC's together simultaneously to to look for a solution to the system of FOC's. If the Profit functions are all concave in their own strategy, then any such solution will be an equilibrium. Don't forget to ensure that the solution is interior. There may be a boundary solution.

Back to the example:

$$\pi_1(q_1, q_2) = \left(50 - \frac{q_1 + q_2}{2}\right) q_1 - (q_1)^2$$

$$\frac{\partial \pi_1}{\partial q_1} = \left(50 - \frac{q_1 + q_2}{2}\right) - \frac{q_1}{2} - 2q_1 = 0$$

SOC:

$$\frac{\partial^2 \pi_1}{\partial q_1^2} = -\frac{1}{2} - \frac{1}{2} - 2 < 0$$

Things are similar for seller 2 by symmetry.

To solve things, we can use each player's FOC as each of these will need to be satisfied at any interior Nash equilibrium. Writing 1's FOC in terms of q_1 as a function of q_2 we obtain what is called 1's best response function:

$$q_1(q_2) = \frac{100 - q_2}{6}.$$

By symmetry, 2's best response function is

$$q_2(q_1) = \frac{100 - q_1}{6}.$$

The Nash equilibria will be solutions to the system where each FOC is solved, that is:

$$q_1 = \frac{100 - q_2}{6}$$

$$q_2 = \frac{100 - q_1}{6}$$

This has a unique solution of $q_1 = q_2 = \frac{100}{7}$ which is interior. So we have a unique Nash equilibrium $(q_1, q_2) = \left(\frac{100}{7}, \frac{100}{7}\right)$.

How can you rule out boundaries? One possibility: Suppose that we are at a boundary, e.g., one of the quantities is zero, say $q_1 = 0$. Then, 2's best response would be to set:

$$q_2 = \frac{100 - q_1}{6} = \frac{100}{6}.$$

But, then 1's best response is:

$$q_1 = \frac{100 - q_2}{6} = \frac{100 - \frac{100}{6}}{6} > 0.$$

Hence, this cannot be a NE.

10.1.1 Generalized Model

Generalize and allow more sellers $n \geq 1$:

$$\pi_i(q, q_i) = (a - bq)q_i - \frac{c}{2}(q_i)^2$$

where $a, b, c > 0$. (Here: $q = \sum_{j=1}^{n} q_j$). FOC is:

$$a - bq) - bq_i - cq_i = 0$$

SOC:

$$\frac{\partial^2 \pi_i}{\partial q_i^2} = -b - b - c < 0$$

Rearrange FOC's to get: $\dfrac{(a - bq)}{(b + c)} = q_i(q)$ (notice they are identical when expressed in terms of q).

Sum these over i to get $q = \dfrac{n(a - bq)}{(b + c)}$. Every interior NE must solve equation.

Solving it for q, you find the unique NE total output to be $q = \dfrac{na}{(n + 1)b + c}$.

Subbing back in to each seller's FOC, we find a unique NE which has each $q_i = \dfrac{a}{(n + 1)b + c}$.

$p = \dfrac{a(b + c)}{(n + 1)b + c}$ and profit $\pi_i = \left[\dfrac{2b + c}{2}\right]\left[\dfrac{a}{(n + 1)b + c}\right]^2$.

Again, boundary solutions can be ruled out, though, as $n \to \infty$, $q_i \to 0$ (boundary).

10.1.2 Long Run

Long Run: Allow entry. Stop at the lowest integer n with positive profit. Long run here is to impose zero profit in a game with entry/exit.

Consider an entry cost $F > 0$.

$$\pi_i = \left[\frac{2b + c}{2}\right]\left[\frac{a}{(n + 1)b + c}\right]^2 - F$$

Find where the profit is zero in the NE of the game with n sellers. This is solved by:

$$n = \frac{1}{b}\left(a\sqrt{\frac{2b+c}{2F}} - 1\right).$$

So, for example, if $a = 100$, $b = 1$, $c = 2$, $F = 9$. Solving the equation we get $n \approx 46.14$. so we choose $n = 46$.

10.2 Other models

- Asymmetric costs can be handled here and the solution will then involve different quantities for each seller.
- Asymmetric, but interrelated demands can be handled, where the goods are not perfect substitutes.
- Asymmetric information can be included, etc.
- Other pricing strategies, e.g., price competition, non-linear tariffs, etc.

They just involve more details and computations. Some varieties are given in the tutorial for this week.

Example 10.2.1 (Asymmetric Cost)**.** A market is made up of a finite set $N = \{1, \cdots, n\}$ of sellers who compete in quantities x_i. They face a demand curve $P(x) = a - bx$ where $a, b > 0$ are parameters and $x = \sum_{i \in N} x_i$. The cost to seller i is $C_i(x_i) = \frac{i}{2}(x_i)^2$. In other words, $C_1(x_1) = \frac{(x_1)^2}{2}$, $C_2(x_2) = (x_i)^2$, etc.

1. Find the set of interior Nash equilibria for each configuration of parameters

   Step 1: Find the profit maximizing output of each seller as a function of the total output x:

   The profit to seller i is:

   $$\pi_i = (a - bx)x_i - i\frac{(x_i)^2}{2}$$

   As the revenue is concave and cost is convex, in x_i, the profit is concave in x_i. Hence SOC's are satisfied and we can focus on FOC's for interior NE.

   $$\frac{\partial \pi_i}{\partial x_i} = (a - bx) - (b + i)x_i = 0$$

   Hence:

   $$x_i(x) = \frac{a - bx}{b + i} \qquad (1)$$

   Step 2: Find the NE quantities assuming an interior solution:

   For an interior NE, it is clear from the above equation (1) that $x < \frac{a}{b}$. Otherwise, each seller would optimally choose to set $x_i = 0$.

   Summing the FOC's we obtain the equation:

   $$x = (a - bx)\left(\sum_{i=1}^{n} \frac{1}{(b + i)}\right)$$

Defining $R \equiv \sum_{i=1}^{n} \frac{1}{(b+i)}$ and simplifying the above expression, we obtain that in every interior NE equilibrium,

$$x = \left(\frac{aR}{1+bR}\right)$$

Since $\frac{aR}{1+bR} < \frac{a}{b}$, it follows that there is a unique NE total output value $x \in (0, \frac{a}{b})$.

Since $x_i(x)$ is a function, and thus assigns a unique value to each $x \in (0, \frac{a}{b})$, there is a unique interior NE.

Plugging this value for x back into $x_i(x)$ and solving we obtain the unique interior NE quantity for each seller as:

$$x_i = \frac{a}{(1+bR)(b+i)} > 0 \text{ for all } i \in N.$$

So far, we have shown that there is a unique interior NE for any configuration of the parameters a, b and n.

2. Could there be a NE where some player chooses $x_i = 0$?

   Let's try to rule out non-interior solutions: Consider any such strategy profile where $x_i = 0$ for some seller (non-interior). Since $x_i = 0$, it follows from the profit function of player i that:

   1. either i is not maximizing profit given the choices of others; or
   2. or $a - bx \leq 0$.

   In the former case, i's NE condition is not satisfied so this cannot be a NE. In the latter case, $x \geq \frac{a}{b} > 0$, so there must be at least one seller j with $x_j > 0$. In this case j is not maximizing, since it would be optimal to set $x_j = 0$

3. We could also consider the long run equilibrium number of sellers with a fixed cost $F > 0$. As each additional seller has a high cost, as F rises the number of sellers falls..

Summary

- Oligopoly has many variations, e.g., price comp, quantity competition, other strategic variables, like quality and advertising
- There is also the possibility to consider decisions over time
- This gives a great variety of potential for analysis of imperfect competition in markets.

11 General Equilibrium Theory

This week we look at General Equilibrium. This allows for many markets simultaneously interacting. It is meant to describe the entire economy. It is a general model of perfect competition.

The term general equilibrium is to capture all markets interacting. Partial equilibrium is what we looked at before with perfect competition and imperfect competition. We will focus on an exchange economy, but the theory is rich enough to include many different production technologies and the implied supplies as well as a variety of preferences and the implied demands across markets. A crucial assumption in the theory is price taking by buyers and sellers.

We focus on a exchange economy where each individual is endowed with some amount of each good. There can be many goods, and many individuals, but let's focus on two goods and two individuals. Let us take Sarah with the following utility function:

$$u_1(x_1, x_2) = x_1^2 x_2 \text{ having an endowment of } \boldsymbol{\omega}^1 = (\omega_1^1, \omega_2^1) = (2, 2).$$

She could consume this endowment, or trade it in competitive markets at the price $\mathbf{p} = (p_1, p_2)$ in the market. We presume she will make her decision about how much to buy and sell of each good to maximize her utility given her wealth.

Sarah's wealth (or income) is implicitly determined by the value of her endowment. In a general equilibrium with production, she could own shares in firms and her wealth would be determined by the combination of her endowment value and her share of the profits from production. Here we consider only exchange, so her income is entirely the value of her endowment, which is

$$p_1\omega_1^1 + p_2\omega_2^1 = 2p_1 + 2p_2.$$

As we can see, her wealth depends on these prices and her endowment. We proceed as before with consumer theory to generate her demand, taking her income as the value of her endowment.

As we know, given Sarah's preferences, $u_1(x_1, x_2) = x_1^2 x_2$, Her demands will be $x_1(\mathbf{p}, I) = \dfrac{2I}{3p_1}$ and $x_2(\mathbf{p}, I) = \dfrac{I}{3p_2}$. Here income I is determined by her endowment. In this example, $I = 2p_1 + 2p_2$. Thus,

$$x_1(\mathbf{p}) = \frac{2(2p_1 + 2p_2)}{3p_1} \quad \text{and} \quad x_2(\mathbf{p}) = \frac{2p_1 + 2p_2}{3p_2}.$$

Observe that as the prices rise, so does her income. On the other hand, a rise in prices simultaneously reduces her purchasing power. In effect, if relative prices don't change, her demand does not change.

$$x_1(\mathbf{p}) = \frac{2(2p_1 + 2p_2)}{3p_1} \quad \text{and} \quad x_2(\mathbf{p}) = \frac{2p_1 + 2p_2}{3p_2}$$

are the final demands of Sarah. It is presumed that she can buy and sell in competitive markets at the prices $\mathbf{p} = (p_1, p_2)$. Let's suppose that the prices are $\mathbf{p} = (1, 2)$. Then, her demands are:

$$x_1 = \frac{2(2+4)}{3} = 4 \quad \text{and} \quad x_2 = \frac{2+4}{6} = 1.$$

Recall that Sarah's endowment is $\boldsymbol{\omega}^1 = (2, 2)$, so: She will purchase an additional 2 units of the first good, and sell 1 unit of the second good.

It is useful to talk of excess demand of good k which is defined as:

$$z_k(\mathbf{p}) = x_k(\mathbf{p}) - \omega_k.$$

This can be positive (when she demands more of some good than her endowment) or negative (when she demands less than her endowment). Think of it as the difference between Sarah's Demand and Supply. At prices $\mathbf{p} = (1, 2)$ with endowment $(2, 2)$ Sarah's excess demand for good 1 is:

$$z_1(1, 2) = 4 - 2 = 2.$$

For good 2, her excess demand is:

$$z_2(1,2) = 1 - 2 = -1.$$

Could both be positive?

In a general equilibrium, we look for prices that lead to supply equal to demand in each market, or equivalently, zero excess demand (over all consumers) in each market.

In this example, if Sarah is the only person in the market, then these prices are not in equilibrium. She would like to buy more of good 1 and less of good 2 than she has, but there is no one offering to sell good 1 or buy good 2. Perhaps, this sounds like a silly exercise, but are there prices that will lead to supply equals demand here, or equivalently, no excess demand in any market? Yes! Sometimes we do this in trade theory (autarky with a representative consumer).

This requires in market 1 that demand equals supply:

$$x_1(\mathbf{p}) = \frac{2(2p_1 + 2p_2)}{3p_1} = 2.$$

There may be multiple solutions to this problem, and in general equilibrium multiplicity happens. One source of multiplicity is the fact that only relative prices matter. To sort out that problem, let's fix one price at a convenient number, say, $p_1 = 1$. Then the problem is to find p_2 solving:

$$\frac{2(2 + 2p_2)}{3} = 2.$$

This is solved when: $p_2 = \frac{1}{2}$. At this price, her demand in market 1 will be exactly her endowment.

This requires in market 2 that:

$$x_2(\mathbf{p}) = \frac{2p_1 + 2p_2}{3p_2} = 2.$$

It happens, in this case that the prices $\mathbf{p} = \left(1, \frac{1}{2}\right)$ satisfy:

$$x_2(\mathbf{p}) = \frac{2+1}{3(0.5)} = 2.$$

In other words, at these prices, supply equals demand in both markets! So, these prices constitute a general equilibrium in this economy. Moreover, the equilibrium is unique up to relative prices. Other equilibria are $(2,1)$, $(4,2)$, ... etc.

Now, we turn to the general theory.

Definition 11.0.1 (Exchange Economy)**.** An exchange economy $e = (G, C, \boldsymbol{\omega}^i, u^i)_{i \in C}$ consists of:

(a) a finite set of goods $G = \{1, \ldots, K\}$;

(b) a finite set of consumers $C = \{1, \ldots, I\}$ where each consumer i has an endowment of goods

$$\boldsymbol{\omega}^i = (\omega_1^i, \ldots, \omega_K^i) \in \mathbb{R}_+^K$$

and a utility function u^i over bundles of goods

$$\mathbf{x}^i = (x_1^i, \ldots, x_K^i) \in \mathbb{R}_+^K.$$

Definition 11.0.2 (Allocation)**.** An allocation is a vector $\mathbf{x} = (\mathbf{x}^1, \ldots, \mathbf{x}^I)$ where each $\mathbf{x}^i = (x_1^i, \ldots, x_K^i) \in \mathbb{R}_+^K$ is a bundle of goods for consumer $i \in C$. (It assigns a bundle to each consumer)

Definition 11.0.3 (Feasible Allocation)**.** An allocation $\mathbf{x} = (\mathbf{x}^1, \ldots, \mathbf{x}^I)$ is called a feasible allocation iff for each $k \in \{1, \ldots, K\}$,

$$\sum_{i=1}^{I} \omega_k^i \geq \sum_{i=1}^{I} x_k^i.$$

(The total endowment is enough to support the allocation)

Definition 11.0.4 (General Equilibrium (Walrasian Equilibrium))**.** A general equilibrium in an exchange economy e consists of a price vector

$$\mathbf{p} = (p_1, \ldots, p_K) \in \mathbb{R}_{++}^K$$

and an allocation $\mathbf{x}$ satisfying:

1. Utility maximization: For each consumer $i \in C$,

$$u^i(\mathbf{x}^i) \geq u^i(\mathbf{y}^i) \quad \text{for all } \mathbf{y}^i \in \mathbb{R}_+^K \text{ satisfying } \sum_{k=1}^{K} p_k y_k^i \leq \sum_{k=1}^{K} p_k \omega_k^i.$$

2. Supply Equals Demand: For each good $k \in \{1, \ldots, K\}$,

$$\omega_k \equiv \sum_{i=1}^{I} \omega_k^i = \sum_{i=1}^{I} x_k^i \equiv x_k.$$

We will put some standard restrictions on preferences and endowments and derive some results that are given in your textbook.

Definition 11.0.5 (Assumption 5.1)**.** $u^i(\mathbf{x})$ represents continuous, strongly monotonic and strictly convex preferences.

Apparent benefits: existence of a unique optimal bundle for any set of prices that is on the budget line.

Theorem 11.0.6. *If assumption 5.1 is satisfied, then: for each consumer i and each price vector $\mathbf{p} \gg 0$, there is a unique optimal bundle. In addition, the demand $x_k^i(\mathbf{p}, \mathbf{p} \cdot \boldsymbol{\omega}^i)$ for each good k is continuous in prices.*

Definition 11.0.7. For each good k, the aggregate excess demand function is the real valued function:

$$z_k(\mathbf{p}) \equiv \sum_{i \in C} x_k^i(\mathbf{p}, \mathbf{p} \cdot \boldsymbol{\omega}^i) - \sum_{i \in C} \omega_k^i.$$

We also define the aggregates:

$$x_k \equiv \sum_{i \in C} x_k^i(\mathbf{p}, \mathbf{p} \cdot \boldsymbol{\omega}^i), \qquad \omega_k \equiv \sum_{i \in C} \omega_k^i, \qquad \mathbf{z}(\mathbf{p}) \equiv \Big(z_1(\mathbf{p}), \ldots, z_K(\mathbf{p})\Big).$$

Note: the aggregates are summed over the individuals, so they represent the Market Demand, Market Supply and Market Excess Demands.

Theorem 11.0.8. *If assumption 5.1 is satisfied, then for all* $\mathbf{p} \gg 0$*:*

1. *(Continuity)* $z_k(\mathbf{p})$ *is continuous for each* k*.*
2. *(Homogeneity of degree zero)* $\mathbf{z}(t\mathbf{p}) = \mathbf{z}(\mathbf{p})$ *for all* $t > 0$*.*
3. *(Walras Law)* $\mathbf{p} \cdot \mathbf{z}(\mathbf{p}) = 0$*.*

Why?

1. Sum of continuous functions is continuous.
2. Demand is homogeneous of degree zero in prices and income.
3. More is better requires each to consume on the budget line.

Theorem 11.0.9. *Let* $\mathbf{x}, \mathbf{p}$ *be a pair of an allocation* $\mathbf{x} = (\mathbf{x}^1, \ldots, \mathbf{x}^I)$ *and a price vector* $\mathbf{p}$ *in an economy* e*. Then,* $(\mathbf{x}, \mathbf{p})$ *is a general equilibrium if and only if excess demand is zero for each good and each consumer's allocation* $\mathbf{x}^i$ *is utility maximizing given* $\mathbf{p}$ *and* $\boldsymbol{\omega}^i$*.*

Theorem 11.0.10. *A general equilibrium exists in an economy* e *whenever assumption 5.1 is satisfied and the total endowment of each good is greater than zero.*

- Existence is quite prevalent.
- This is a sufficient condition.
- Existence may happen with weaker conditions, you just need to check for it.

Recall from Theorem 5.2 that under the assumptions of Theorem 5.5, we get:

1. Uniqueness and continuity of demand at each price vector $\mathbf{p}$ for each good from each consumer and as a consequence in the market.
2. Excess Demand is homogeneous of degree zero in prices:

$$z_k(t\mathbf{p}) = z_k(\mathbf{p}) \quad \text{for all } t > 0.$$

3. Walras Law: value of excess demand is zero at any prices.

$$\sum_{k=1}^{K} p_k z_k(\mathbf{p}) = 0 \quad \text{for all } \mathbf{p} \in \mathbb{R}^K_{++},$$

where:

$$z_k(\mathbf{p}) \equiv \sum_{i=1}^{I} z_k^i(\mathbf{p}) \quad \text{(sum of excess demands over consumers).}$$

The properties listed on the previous slide help to prove existence which typically is done using a fixed-point theorem. The properties 2 and 3 also help us to simplify the process of finding or computing a general equilibrium (when it exists).

- Property 2 (Homogeneity of Degree zero in prices) implies that we can fix a price at 1 and just consider relative prices, since that is all that matters. This actually is just a result of utility maximization and needs no further assumptions.
- Property 3 (Walras' Law) implies that if supply equals demand in all but one market, then supply equals demand in the remaining market. This needs only some form of monotonicity of preferences to get consumers to the budget line.

It means we can drop one market in looking for an equilibrium. If the prices clear $K-1$ markets, then the last market also clears. Note our result in the autarky market with one consumer and 2 goods. We found that when market 1 clears, so does market 2.

Let's add John to Sarah's economy, where John has the utility function

$$u_2(x_1, x_2) = x_1 x_2$$

and endowment $\boldsymbol{\omega}^2 = (2, 3)$.

John's demands are:

$$x_1^2 = \frac{I}{2p_1} = \frac{2p_1 + 3p_2}{2p_1} \quad \text{and} \quad x_2^2 = \frac{I}{2p_2} = \frac{2p_1 + 3p_2}{2p_2}.$$

Sarah's demands are:

$$x_1^1 = \frac{4(p_1 + p_2)}{3p_1} \quad \text{and} \quad x_2^1 = \frac{2(p_1 + p_2)}{3p_2}.$$

Noting Properties 2 and 3 hold, we set $p_1 = 1$ and just focus on market 1. This leads to the requirement of:

$$\text{(demand)} \quad \frac{4(1 + p_2)}{3} + \frac{2 + 3p_2}{2} = 2 + 2 \quad \text{(supply).}$$

This is solved when:

$$p_2 = \frac{10}{17}.$$

The unique general equilibrium (up to relative prices) is:

$$\mathbf{p} = \left(1, \frac{10}{17}\right)$$

with allocation: Sarah $\mathbf{x}^1 = \left(\frac{36}{17}, 1.8\right)$ and for John $\mathbf{x}^2 = \left(\frac{32}{17}, 3.2\right)$.

Using the price vector, we can find Sarah's demand for good 1 and good 2. Noting the allocations must sum to supply in each market, John gets the rest. It is no harm to double check that these quantities are demanded by each person.

The previous example did not satisfy all the assumptions for Theorem 5.5, yet there was an equilibrium. Sometimes, the violation of those assumptions will create a situation where there is no competitive equilibrium. The only way to know for sure is to check. Also, sometimes we can get multiple general equilibria.

Example 11.0.11 (No equilibrium.)**.** Lisa has $u_1 = \max\{x_1, x_2\}$ with endowment $(1, 2)$. Jeff has $u_2 = \min\{x_1, x_2\}$ with endowment $(2, 1)$. Fixing $p_1 = 1$ we find Lisa will have the following demands for good 1:

$$x_1^1 = \begin{cases} 1 + 2p_2 & \text{if } p_2 > 1, \\ 0 \text{ or } 3 & \text{if } p_2 = 1, \\ 0 & \text{if } p_2 < 1. \end{cases}$$

Jeff's demand for good 1 is:

$$x_1^2 = \frac{2 + p_2}{1 + p_2}.$$

Given Lisa's demand, we have 3 cases: (a) $p_2 > 1$, (b) $p_2 = 1$; (c) $p_2 < 1$.

Checking:

- Case (a) $p_2 > 1$: (demand)

$$1 + 2p_2 + \frac{2 + p_2}{1 + p_2} > 3 \quad \text{(supply)}$$

  since $p_2 > 1$. Hence, no equilibrium.

- Case (b) $p_2 = 1$:

$$\text{either} \quad 0 + \frac{2 + 1}{1 + 1} = 1.5 < 3 \quad \text{(supply)}$$

  or

$$3 + \frac{2 + 1}{1 + 1} = 4.5 > 3 \quad \text{(supply)}.$$

  Hence, no equilibrium.

- Case (c) $p_2 < 1$: (demand) is

$$0 + \frac{2 + p_2}{1 + p_2} < 3 \quad \text{(supply)}$$

  as demand is decreasing in p_2 and thus always less than 2. Hence, no equilibrium in case (c).

All cases exhausted, so no equilibrium.

Example 11.0.12 (Multiple Equilibria Example)**.** Lisa has $u_1 = x_1 + 3.5x_2 - 0.5x_2^2$ with endowment $(4, 0)$. Jeff has $u_2 = 3.5x_1 + x_2 - 0.5x_1^2$ with endowment $(0, 4)$. Notice that these are both quasi-linear utilities but with respect to different goods. Again, setting $p_1 = 1$ but now looking at the market for good 2, we find:

$$\text{(demand)} \quad 3.5 - p_2 + 4 - \frac{3.5}{p_2} + \frac{1}{p_2^2} = 4 \quad \text{(supply)}.$$

This has 3 solutions: $p_2 = 0.5$, $p_2 = 1$, $p_2 = 2$. Each leads to a different allocation, and Jeff and Lisa do not rank them the same way.

This example appears in a paper "Simple Economies with Multiple Equilibria" by Bergstrom, Shimomura and Yamamoto (B.E. Journal of Theoretical Economics (2009)). It is a simplification of an example by Shapley-Shubik (1977) Journal of Political Economy).

Summary

- General Equilibrium is a powerful tool for studying interactions across many markets.
- Today we looked at exchange economies.
- The theory extends to production economies where individuals consume and provide resources to production processes and own the means of production.
- We explored some general positive results about general equilibrium.
- Next week we will look at some normative results relating to welfare from general equilibrium.

12 Welfare

Last Week we looked at general equilibrium theory focusing on positive results. This week we focus on normative results in particular welfare.

Definition 12.0.1 (Pareto Efficient)**.** An allocation $\mathbf{x}$ is Pareto efficient in an economy e iff $\mathbf{x}$ is feasible and there is no other feasible allocation $\mathbf{y}$ such that $u^i(\mathbf{y}^i) \geq u^i(\mathbf{x}^i)$ for all consumers $i \in C$ and $u^j(\mathbf{y}^j) > u^j(\mathbf{x}^j)$ for some consumer $j \in C$.

This is the standard definition used in economics for welfare. It is quite weak and allows for very inequitable outcomes. Nevertheless it is a starting point.

Note that logically, the definition says:

An allocation $\mathbf{x}$ is Pareto efficient iff

(a) $\mathbf{x}$ is feasible;

(b) there is no other feasible allocation $\mathbf{y}$ that makes every consumer at least as well off and someone better off.

Part (b) is logically equivalent to "every other allocation $\mathbf{y}$ that makes every consumer at least as well off and someone better off is NOT FEASIBLE." Thus to show a feasible allocation $\mathbf{x}$ is NOT Pareto efficient it suffices to show SOME other feasible allocation $\mathbf{y}$ makes every consumer at least as well off and someone better off. Showing a feasible allocation is Pareto efficient is more difficult.

Example 12.0.2 (2 person exchange economy)**.** $u(x_1, x_2) = x_1 x_2$ and $v(y_1, y_2) = y_1^2 y_2$. Here $x = (x_1, x_2)$ and $u(x_1, x_2)$ denote the bundle and utility of person 1, let's call her Anne. Her endowment is $\omega_A = (5, 1)$. Similarly $y = (y_1, y_2)$ and $v(y_1, y_2)$ denote the bundle and utility of person 2, let's call him Brad. His endowment is $\omega_B = (5, 4)$.

Both have Cobb-Douglas utility functions, but different preferences with person Brad favouring the first good. Let's look for the set of Pareto efficient allocations.

Is the endowment $\omega_A = (5, 1)$, $\omega_B = (5, 4)$ efficient? It is clearly feasible, so (a) is satisfied.

To show (b) is satisfied, we need to show there is no other feasible allocation that makes both at least as well off and someone better off. The set of feasible allocations is very large. It is the set of bundles:

$$F = \left\{(x, y) \in \mathbb{R}^2_+ \times \mathbb{R}^2_+ : \ x_1 + y_1 \leq 10 \text{ and } x_2 + y_2 \leq 5\right\}.$$

In words: it is the set of bundles, one for each person, such that each person gets a non-negative amount of each good and the sum total of each good allocated does not exceed the endowment of each good in the economy.

The set F is convex. You should be able to show this. The endowment gives Anne (person 1) a utility of $u(5,1) = 5 \cdot 1 = 5$ and it gives Brad (person 2) a utility of $v(5,4) = 5^2 \cdot 4 = 100$. Noting that Brad prefers good 1 more, we might imagine a gain if he trades one unit of good 2 for a unit of good 1. This leads to the allocation $(4,2)$ for Anne and $(6,3)$ for Brad. This yields Anne utility of $8 > 5$ and Brad a utility of $108 > 100$. Both are better off, so the endowment is not efficient.

We can use some observations to find the Pareto Efficient Set.

1. As the utility functions are both strictly increasing functions, every efficient allocation will have:

$$x_1 + y_1 = 10 \text{ (good 1)} \quad \text{and} \quad x_2 + y_2 = 5 \text{ (good 2), (NO WASTE).}$$

2. As the utility functions are continuously differentiable and strictly convex at interior solutions, the Marginal Rates of Substitution must be equal for the two consumers. (MRS EQUALIZATION)

Let's explore (2) a bit more. Define the marginal utility $u_i = \dfrac{\partial u}{\partial x_i}$ for good $i = 1, 2$. Recall that the marginal rate of substitution of good 2 for good 1 is defined as:

$$\text{MRS of good 2 for good 1} = \frac{u_1}{u_2}.$$

It measures how much of good 2 a person is willing to give up for another unit of good 1. So, for example, if $u_1 = 5$ and $u_2 = 1$, then 1 unit of good 1 is worth 5 times what a unit of good 2 is worth. Thus, the person is willing to give up 5 units (and no more) of good 2 for a unit of good 1.

On the flip, that person is willing to accept 5 units (and no less) of good 2 for giving up a unit of good 1. At the endowment, we have MRS's of Anne and Brad:

$$\text{MRS of good 2 for good 1 (Anne)} = \frac{u_1(5,1)}{u_2(5,1)} = \frac{x_2}{x_1} = \frac{1}{5} < \frac{8}{5}$$

$$= \frac{2y_2}{y_1} = \frac{v_1(5,4)}{v_2(5,4)} = \text{MRS of good 2 for good 1 (Brad).}$$

This tells us that Brad is willing to give up more units $\frac{8}{5}$ of good 2 for a unit of good 1 than what Anne is willing to accept $\frac{1}{5}$ for giving up a unit of good 2. If, for example, Brad trades a unit of good 2 for a unit of good 1, they will both be better off. This is what we observed previously.

MRS Equalization:

$$(3) \quad \frac{x_2}{x_1} = \frac{2y_2}{y_1}$$

No Waste:

$$(1) \quad x_1 + y_1 = 10 \quad \text{and} \quad (2) \quad x_2 + y_2 = 5$$

We are looking for solutions to (1), (2) and (3).

Solving (1) for y_1 and (2) for y_2 and subbing into the RHS of (3) we get:

$$\frac{x_2}{x_1} = \frac{2(5 - x_2)}{(10 - x_1)}.$$

To obtain a PO allocation:

(a) Fix $x_1 \in (0, 10)$.

(b) Solve for x_2 as a function of x_1 using (4) to get

$$x_2(x_1) = \frac{10x_1}{10 + x_1}.$$

(c) Set $y_1 = 10 - x_1$ and $y_2 = 5 - x_2(x_1)$

Rearranging, we obtain:

$$(4) \quad 10x_1 - x_1x_2 - 10x_2 = 0.$$

This process yields the set of Pareto efficient allocations as:

$$PO = \left\{(x, y) : 0 \leq x_1 \leq 10,\ x_2 = \frac{10x_1}{10 + x_1},\ y_1 = 10 - x_1,\ y_2 = 5 - x_2\right\}.$$

In words, an allocation is Pareto efficient if and only if

- Anne is consuming a bundle $\left(x_1, \dfrac{10x_1}{10 + x_1}\right)$ where $0 \leq x_1 \leq 10$
- Brad is consuming the corresponding bundle $\left(10 - x_1,\ 5 - \dfrac{10x_1}{10 + x_1}\right)$.

Note that PO includes the allocations:

(a) Anne consumes all of both goods: $(x, y) = ((10, 5), (0, 0))$

(b) Brad consumes all of both goods: $(x, y) = ((0, 0), (10, 5))$

The best for one and worst for the other are PO.

General Cases: Finding the Pareto Optimal set of outcomes is not simple but doable.

If all the utility functions in the economy are concave in addition to the other standard conditions, then the set can be found by maximizing the weighted sum of the utilities of the consumers and adjusting the weights. (See Propositions 16.E.2 and 16.F.1 of Microeconomic Theory 1995, by Mas-Colell, Whinston, Green)

We could have used this for the example we are exploring. Even though the utility functions are not concave, they have preferences that can be represented by a concave utility function (e.g., log or make exponents sum to 1 or less than 1).

Now we consider some normative properties of General Equilibria

Theorem 12.0.3 (First Fundamental Theorem of Welfare Economics)**.** *Let e be an economy in which every consumer has monotonic preferences. Every general equilibrium allocation* $\mathbf{x}$ *is Pareto efficient.*

- *This result requires only that consumer demands are always on the budget line.*
- *If they are not, then a competitive equilibrium may not be Pareto efficient.*
- *If a competitive equilibrium does not exist, then it says nothing about the economy.*

Definition 12.0.4 (Blocking)**.** A subset of individuals $S \subseteq C$ in an exchange economy e is said to block an allocation $\mathbf{x}$ iff there is some other allocation $\mathbf{y}$ satisfying:

1. $\sum_{i \in S} \mathbf{y}^i \leq \sum_{i \in S} \boldsymbol{\omega}^i$ (feasible for S)
2. $u^i(\mathbf{y}^i) \geq u^i(\mathbf{x}^i)$ for all $i \in S$ with at least one strict inequality (preferred by S)

Definition 12.0.5 (Core)**.** The core of an exchange economy e is the set of allocations that are not blocked by any subset $S \subseteq C$.

- Every allocation that is in the CORE is Pareto Optimal.
- The subset C would block it if it was not Pareto Optimal.
- CORE is stronger. It requires no other subset would block.

Theorem 12.0.6 (Core Property of General Equilibrium)**.** *Let e be an economy in which every consumer has monotonic preferences. Every general equilibrium allocation $\mathbf{x}$ is in the core.*

This result requires only that consumer demands are always on the budget line. If they are not, then a competitive equilibrium may not be in the core. If a competitive equilibrium does not exist, then it says nothing about the economy.

Finding the CORE: In a pure exchange economy with 2 people, it requires simply that each gets a utility of at least what is given by his endowment and that it is Pareto Efficient. With more than 2 people, it requires more. It requires that each subset of people get more than they can get alone as a subset.

The core concept is used in cooperative game theory: a side of game theory where players act cooperatively to achieve a desirable outcome and split any surplus in a mutually agreeable manner.

In our previous example, we can take the set of Pareto Optimal allocations:

$$PO = \left\{ (x, y) : 0 \leq x_1 \leq 10,\ x_2 = \frac{10x_1}{10 + x_1},\ y_1 = 10 - x_1,\ y_2 = 5 - x_2 \right\}.$$

Then require that each gets at least the utility of his endowment.

- For Anne: $u(x_1, x_2) \geq u(5, 1) = 5$
- For Brad: $v(y_1, y_2) \geq v(5, 4) = 100$

For each PO allocation, Anne does not block if:

$$u(x) = x_1 \left(\frac{10x_1}{10 + x_1} \right) \geq 5$$

This requires:

$$x_1^2 - 0.5x_1 - 5 \geq 0$$

Which requires: $x_1 \geq 2.5$.

For each PO allocation, Brad does not block if:

$$v(y) = y_1^2 \left(\frac{5y_1}{20 - y_1} \right) \geq 100$$

(Here, I put things in terms of y to ease my calculations)

This requires:

$$y_1^3 \geq 20(20 - y_1)$$

In other words:

$$y_1^3 + 20y_1 - 400 \geq 0$$

This requires: $y_1 \geq 6.47$ (approximately). As $y_1 + x_1 = 10$, this translates to $x_1 \leq 3.53$.

$$\text{CORE} = \left\{(x, y) : 2.5 \leq x_1 \leq 3.53,\ x_2 = \frac{10x_1}{10 + x_1},\ y_1 = 10 - x_1,\ y_2 = 5 - x_2\right\}.$$

Fix price of good 1 at 1 and let p denote the price of good 2. Then Anne's demand for good 1 is:

$$x_1 = \frac{5 + p}{2}$$

Brad's demand for good 1 is:

$$y_1 = \frac{2(5 + 4p)}{3}$$

Demand equals supply for good 1:

$$\frac{5 + p}{2} + \frac{2(5 + 4p)}{3} = 10$$

So,

$$p = \frac{25}{19} \approx 1.32$$

So $x_1 = 3.16$ which puts us in the Core.

$$\text{CORE} = \left\{(x, y) : 2.5 \leq x_1 \leq 3.53,\ x_2 = \frac{10x_1}{10 + x_1},\ y_1 = 10 - x_1,\ y_2 = 5 - x_2\right\}.$$

Example 12.0.7 (Example with 3 consumers)**.** $u_i(x_1, x_2) = x_1x_2$ for each consumer (just like Anne). Endowments are $(19, 1)$, $(1, 19)$ and $(10, 10)$ with utilities $19, 19, 100$.

The allocation $(9, 9)$, $(9, 9)$, $(12, 12)$ is Pareto Optimal and makes each better than with the endowment. $81 > 19$ and $144 > 100$. However, the first two will block this allocation using $(10, 10)$ and $(10, 10)$ which is feasible for them. Hence, the allocation $(9, 9)$, $(9, 9)$, $(12, 12)$ is not in the Core.

Theorem 12.0.8 (Second Fundamental Theorem of Welfare Economics)**.** *Let e be an economy in which assumption 5.1 is satisfied and the total endowment of each good is greater than zero. Every Pareto efficient allocation $\mathbf{x}$ can be obtained by appropriately redistributing the endowments of the consumers.*

Let $\omega = (\omega^i)_{i \in C}$ be the initial endowment and $\mathbf{x} = (\mathbf{x}^i)_{i \in C}$ be the desired Pareto efficient allocation. Redistribute endowments from ω to ω' such that $\omega' = \mathbf{x}$. The new endowment is a General Equilibrium.

Both welfare theorems hold for production economies with sufficient conditions on production technologies for the 2nd theorem. The second welfare theorem is less trivial there. It gives some sense that policy can guide us to efficiency by an appropriate tax and transfer system.

Limitations

- Both welfare theorems have some difficulties.
- 1. The first welfare theorem, implicitly assumes there are no externalities (utility depends only on my consumption bundle, nothing else). If there are externalities, general equilibria may not be efficient.
- 2. The first welfare theorem, implicitly assumes price information is readily available. If information is limited, general equilibria may not be Pareto efficient.
- 3. General equilibrium has problems when technologies are not convex, so equilibria may not exist. Then alternative equilibria, e.g., Nash equilibria may be more appropriate. These may not be Pareto efficient.
- The second welfare theorem also has some (perhaps more) difficulties.
- 1. Though governments have the ability to tax and make transfers, this is not perfect. So we may not get to a Pareto efficient outcome of our liking even if markets are competitive.
- 2. Information is scarce and this theorem relies on knowing the preferences in order to find the Pareto efficient allocations. Governments and even individuals may not know their own preferences, let alone those of others. (Notice that the first fundamental welfare theorem does not require knowing this.)
- 3. Individual incentives are ignored in general equilibrium theory. Essentially, if the government is going to redistribute wealth, why should an individual work hard to gain profit?